

Measurement of π^0 and η mesons in oxygen-oxygen collisions at
 $\sqrt{s_{\text{NN}}} = 5.36$ TeV using Machine Learning-based photon
identification in ALICE Run 3

Supplemental figures

Machine-learning training features in proton-proton collisions at
 $\sqrt{s} = 5.36$ TeV

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ML models by Dr. Ana Marin

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1 Feature map

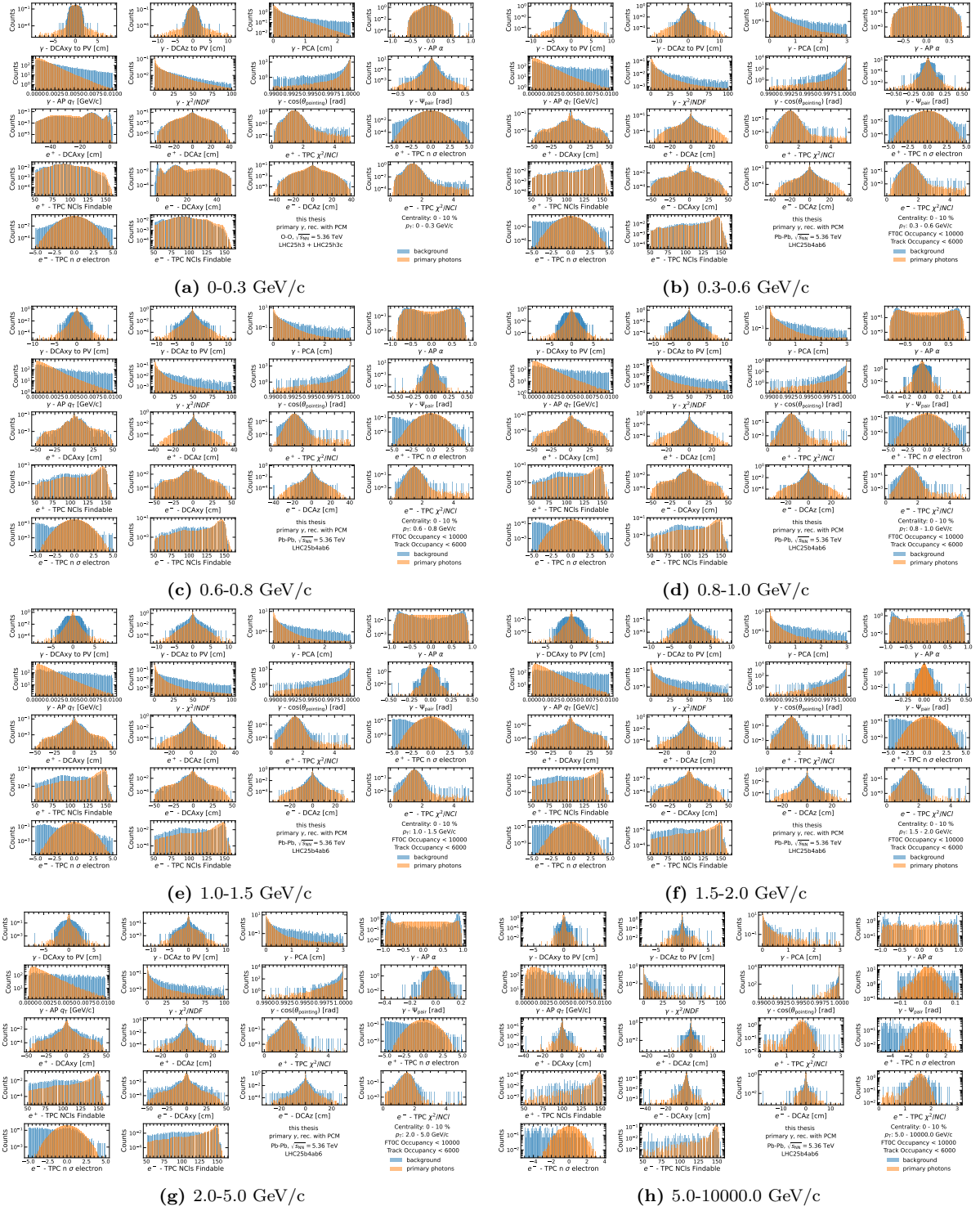


Figure 1: Feature map for all p_T bins in the centrality range 0-10 %

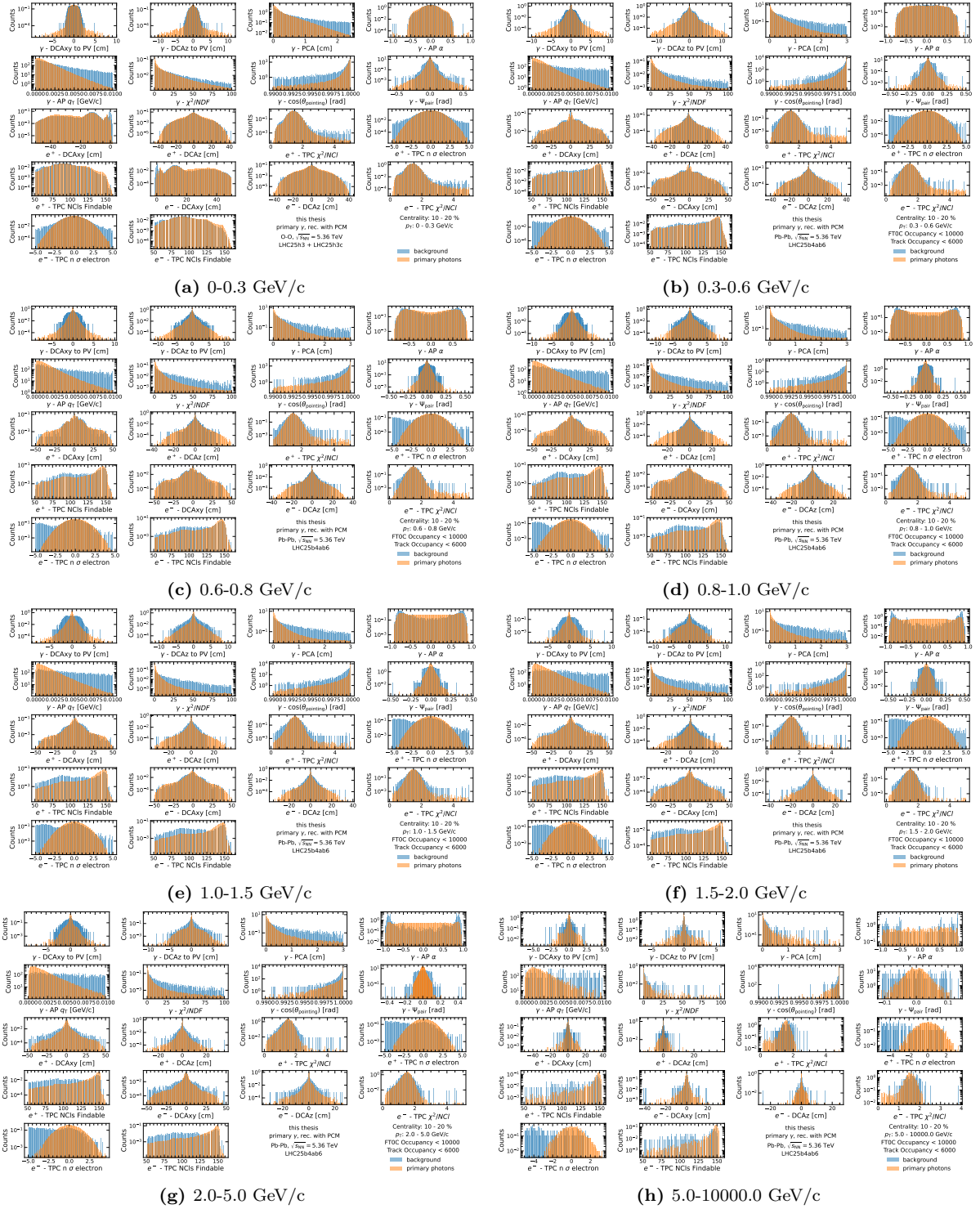


Figure 2: Feature map for all p_T bins in the centrality range 10-20 %

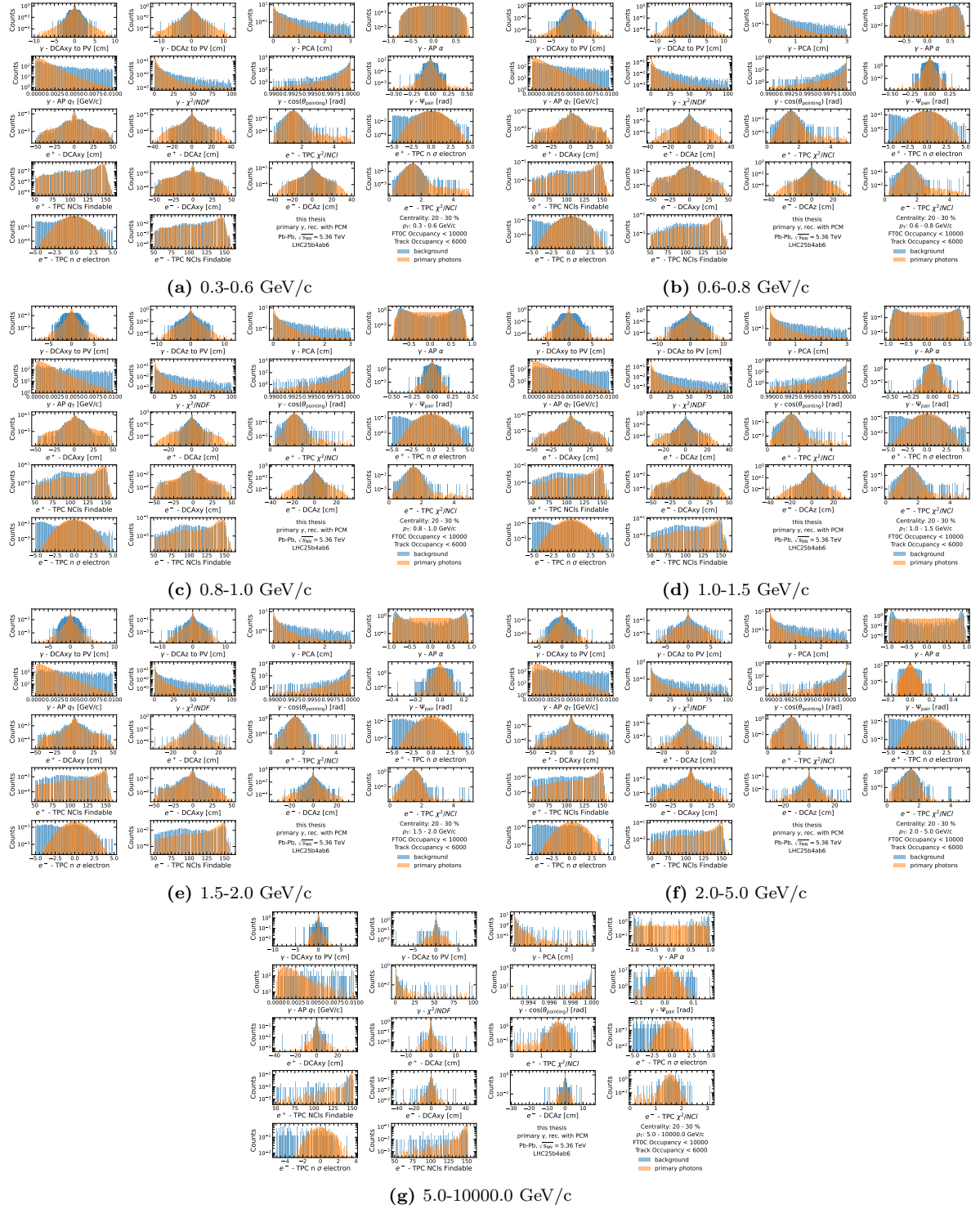


Figure 3: Feature map for all p_T bins in the centrality range 20-30 %

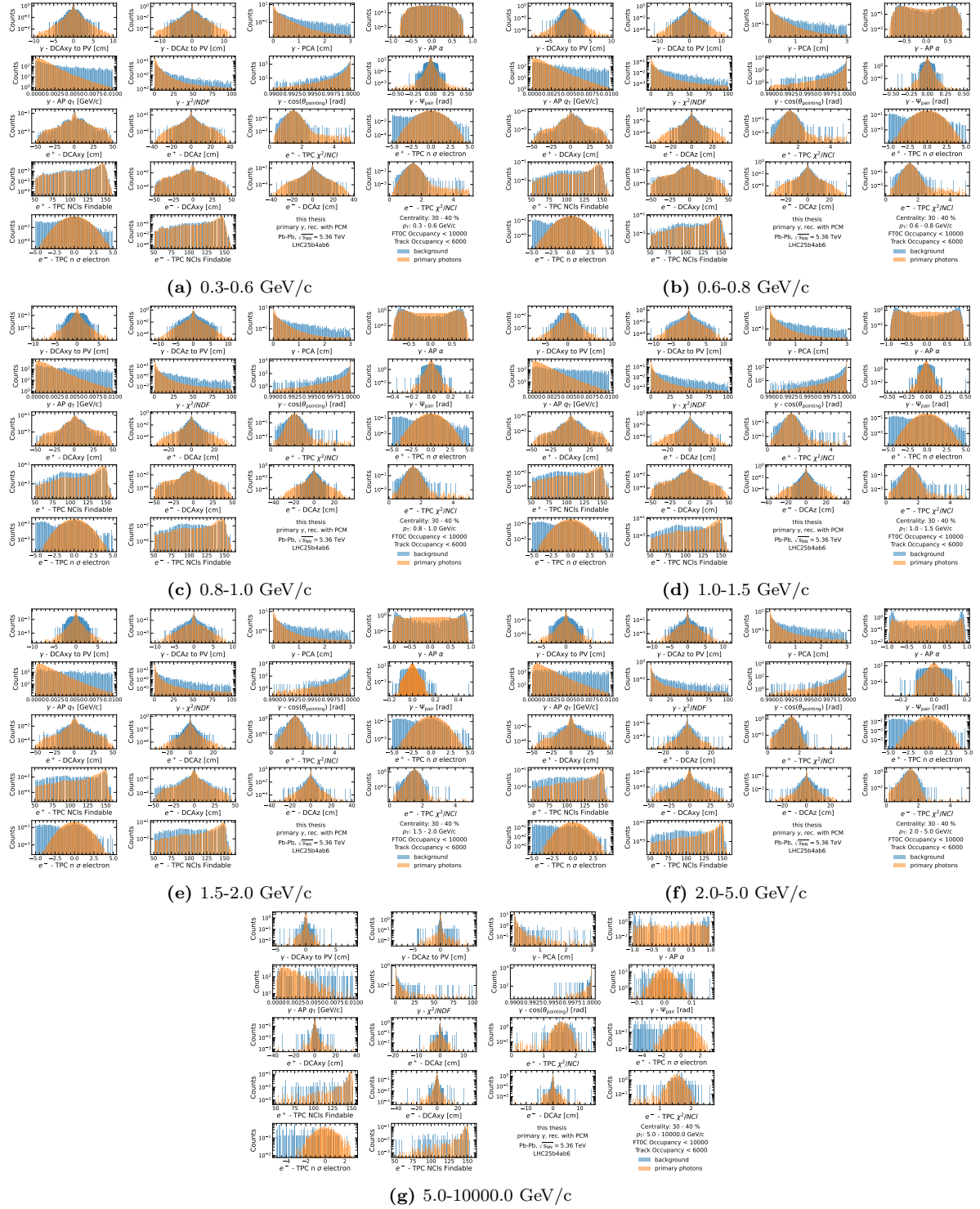


Figure 4: Feature map for all p_T bins in the centrality range 30-40 %

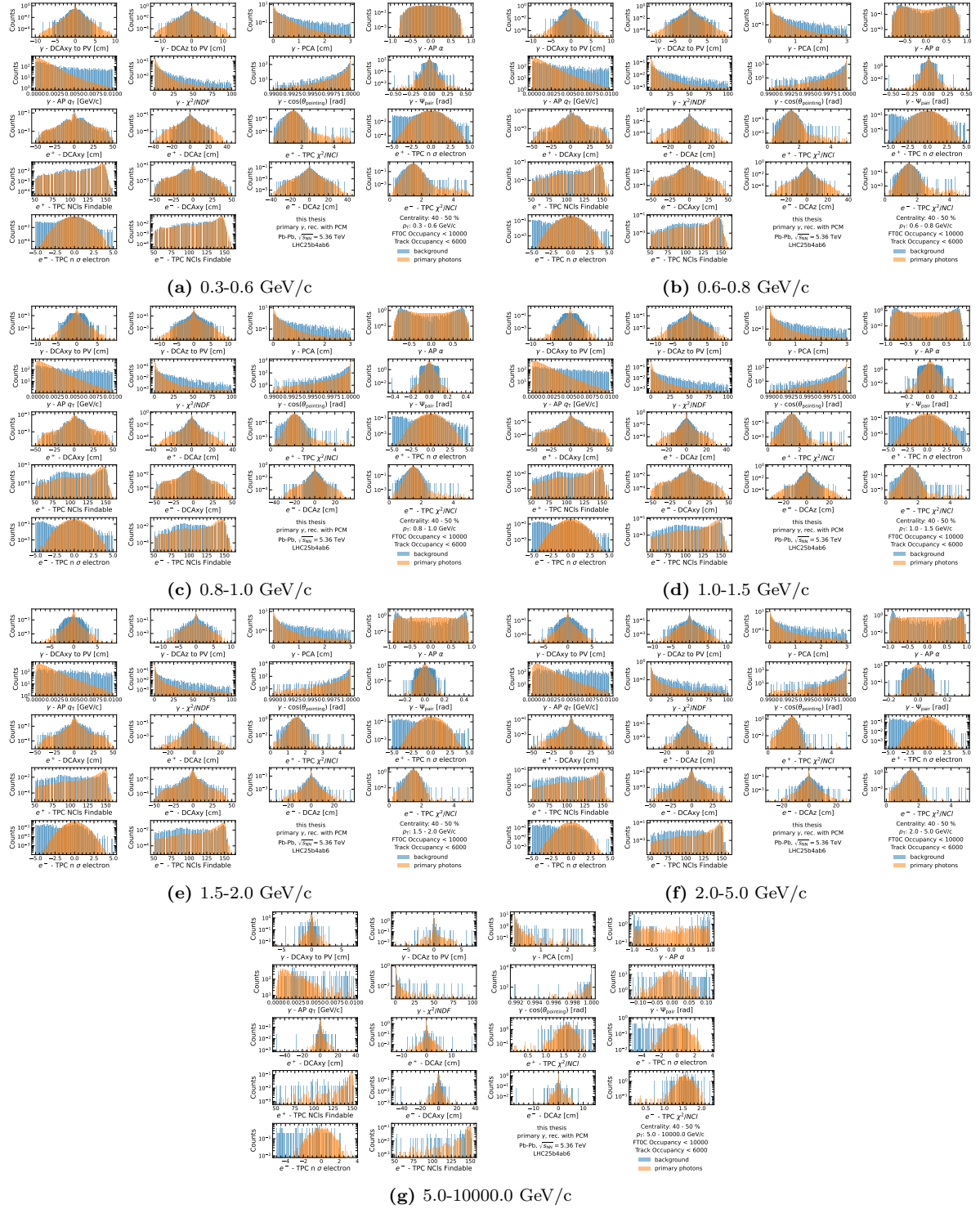


Figure 5: Feature map for all p_T bins in the centrality range 40-50 %

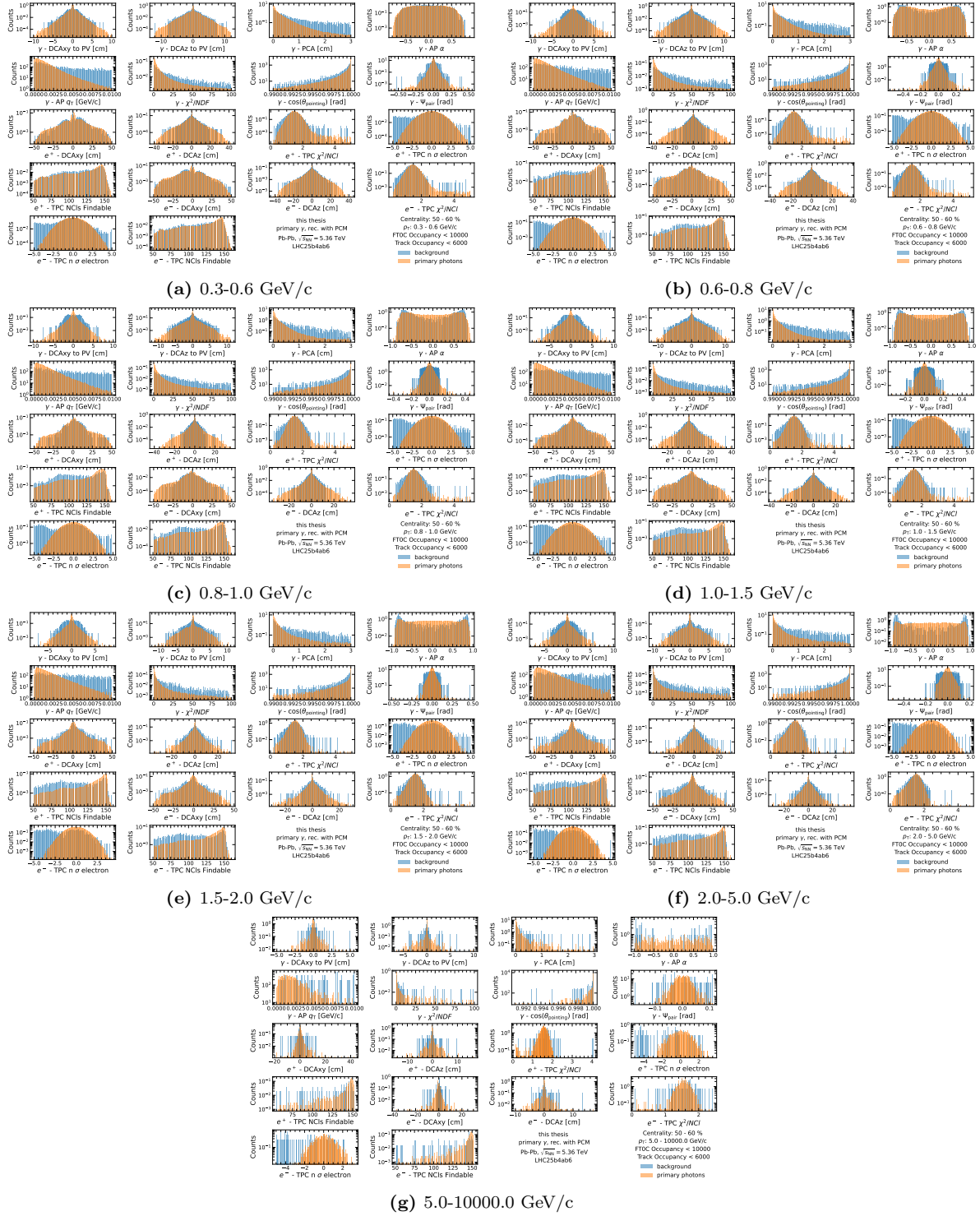


Figure 6: Feature map for all p_T bins in the centrality range 50-60 %

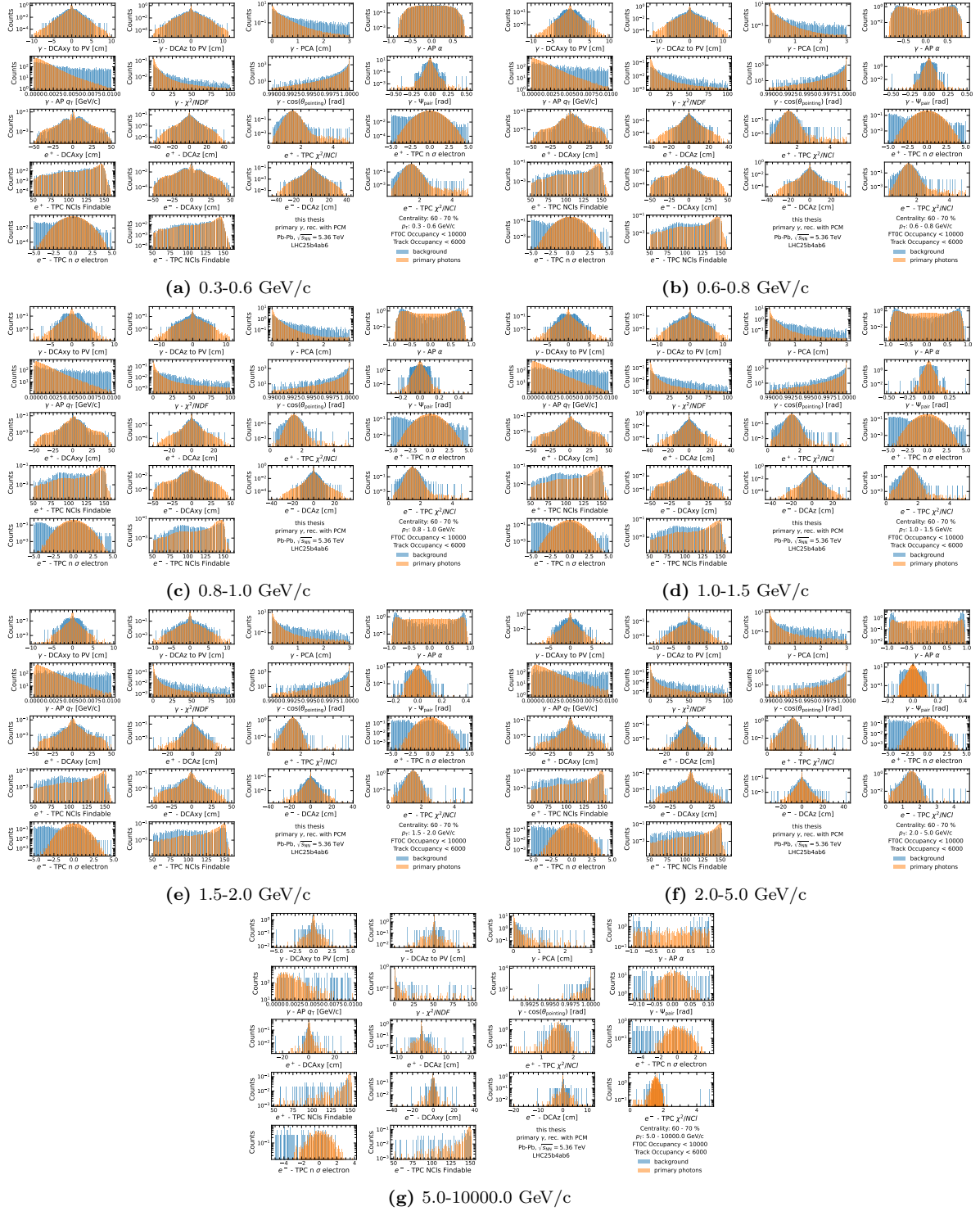


Figure 7: Feature map for all p_T bins in the centrality range 60-70 %

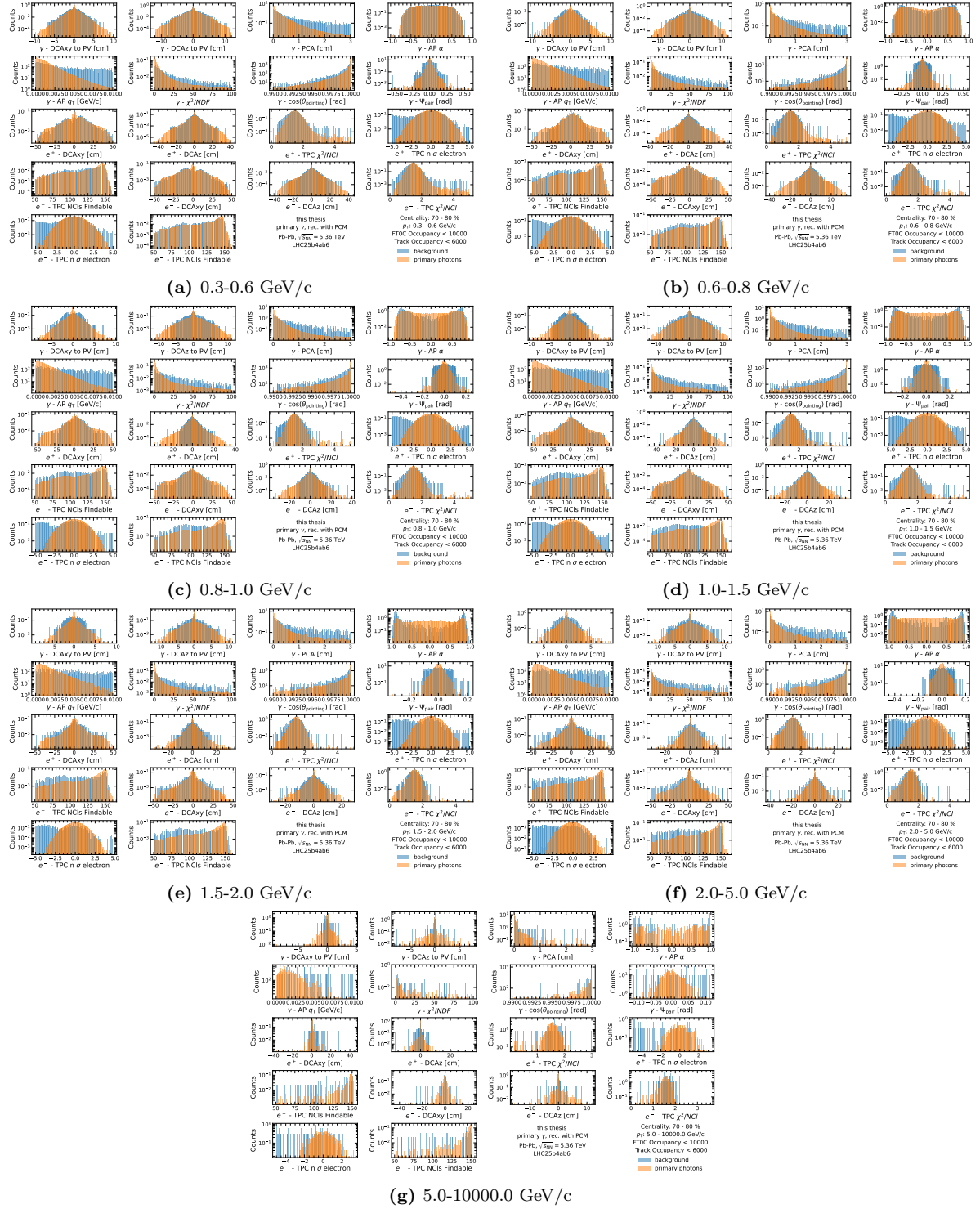


Figure 8: Feature map for all p_T bins in the centrality range 70-80 %

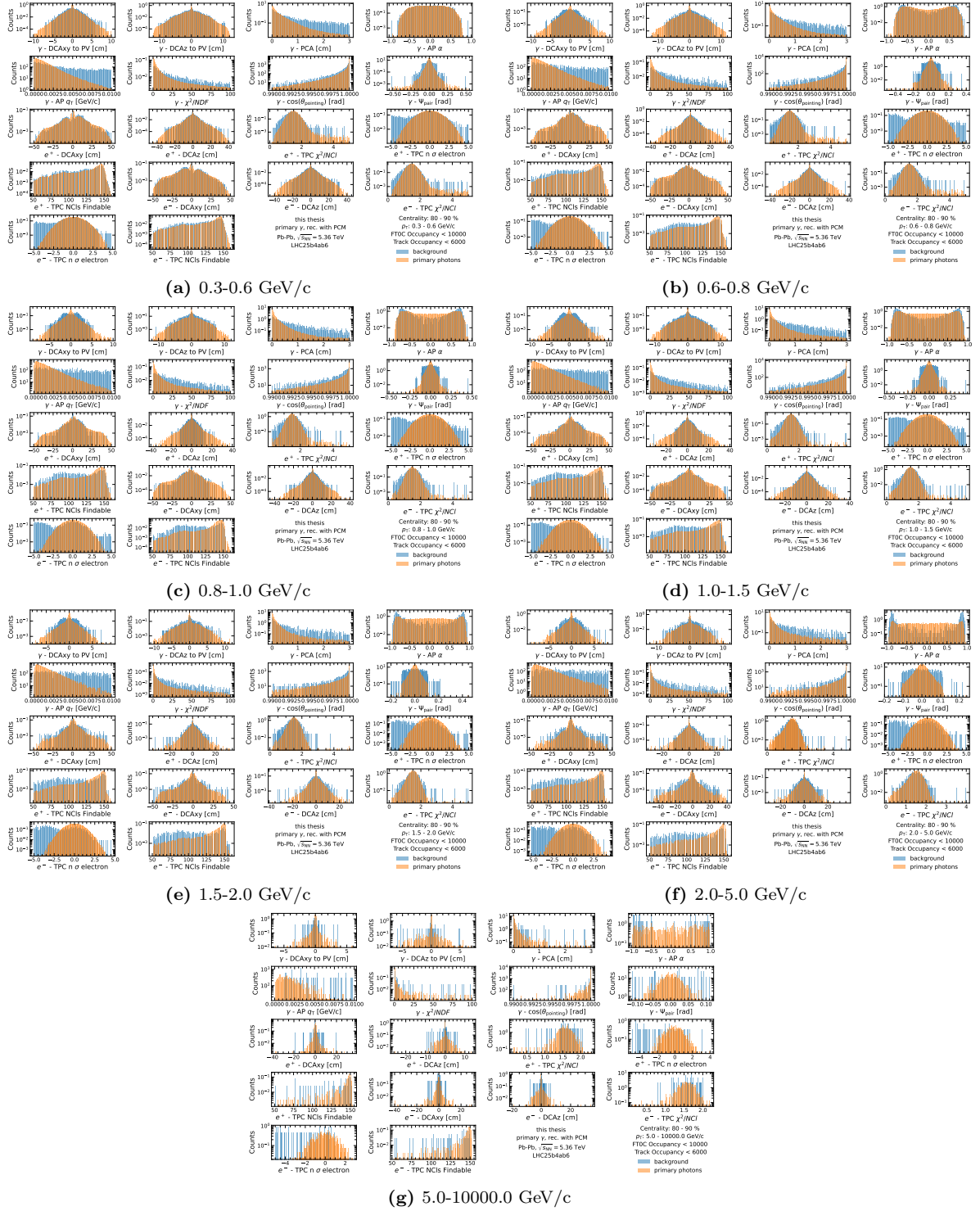


Figure 9: Feature map for all p_T bins in the centrality range 80-90 %

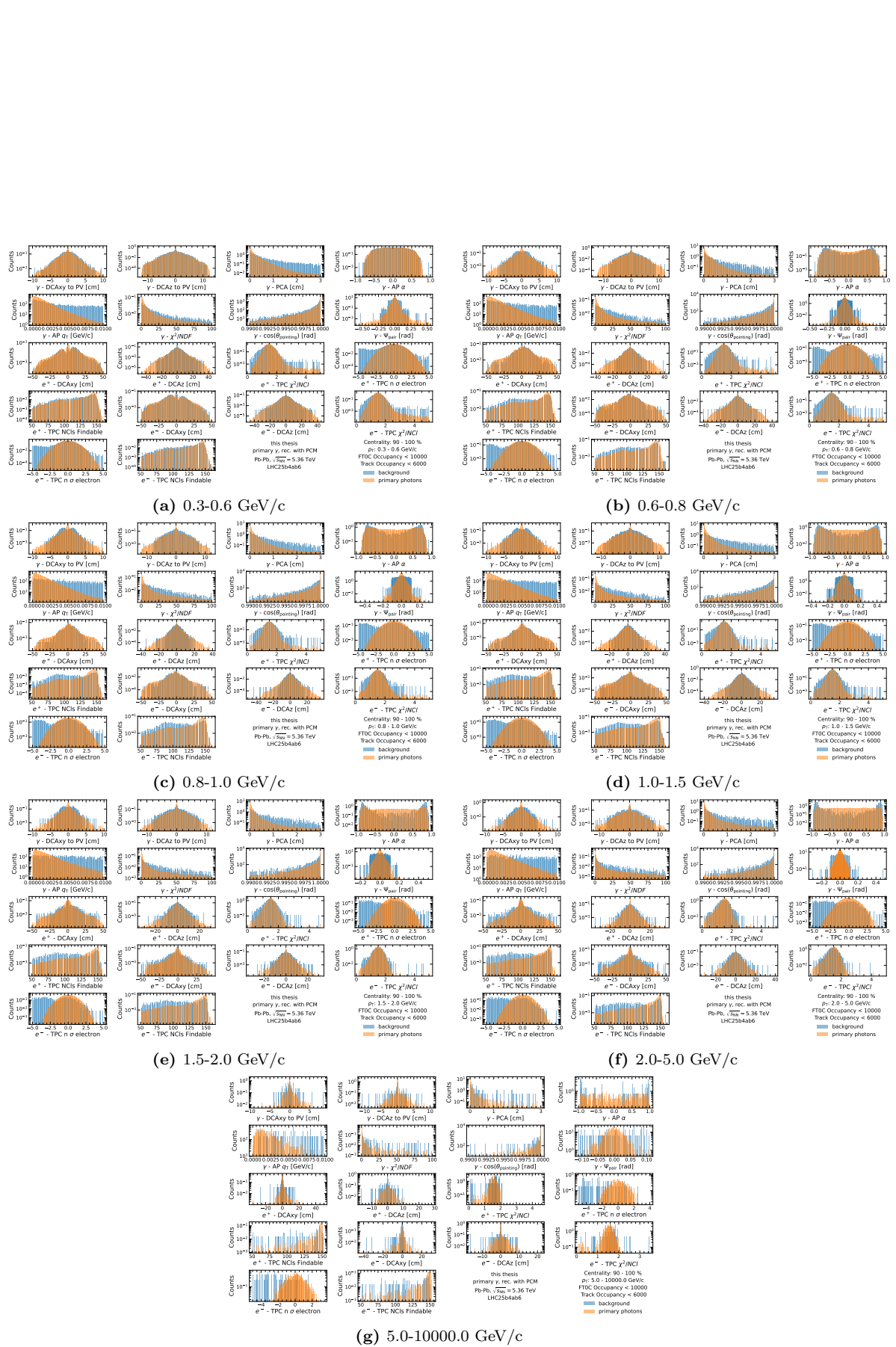


Figure 10: Feature map for all p_T bins in the centrality range 90-100 %

2 Importance of the input features

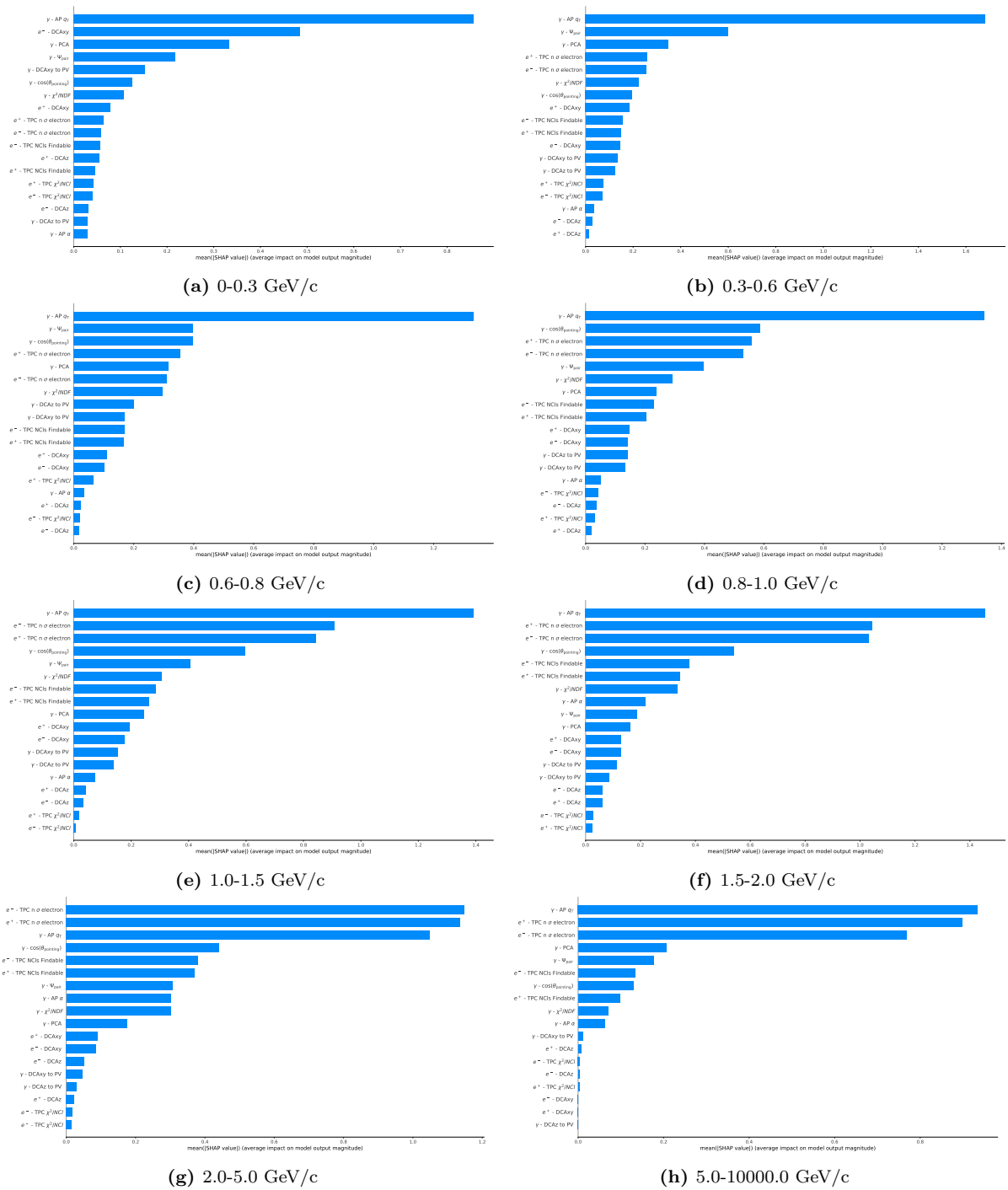
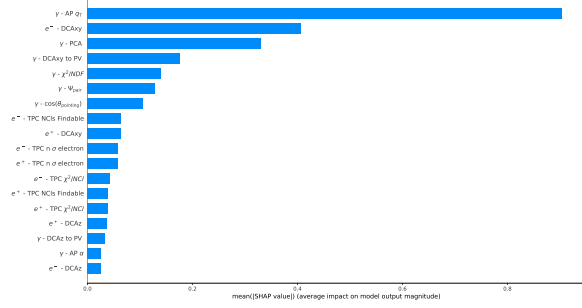
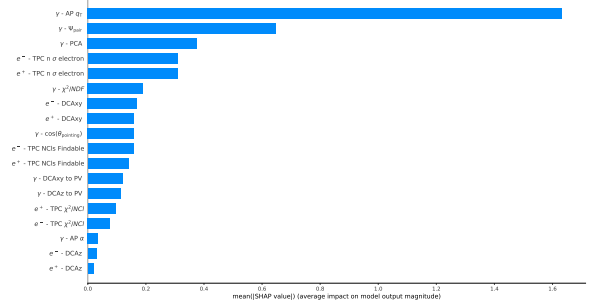


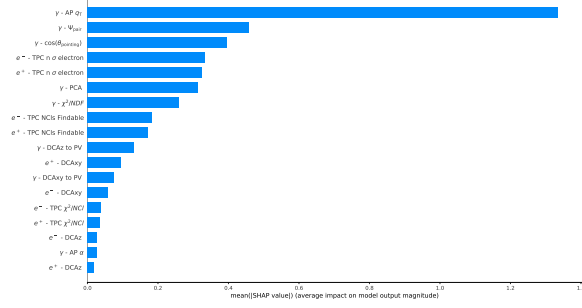
Figure 11: Importance of the input features for all p_T bins in the centrality range 0-10 %



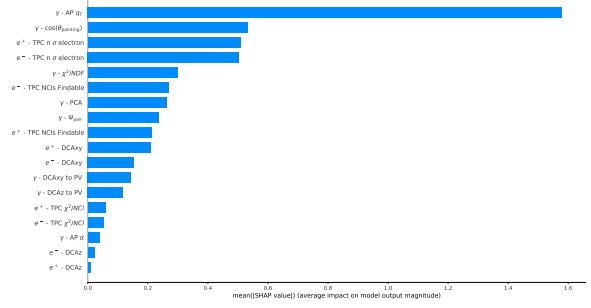
(a) 0-0.3 GeV/c



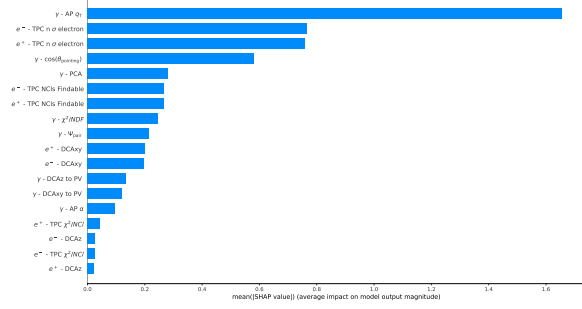
(b) 0.3-0.6 GeV/c



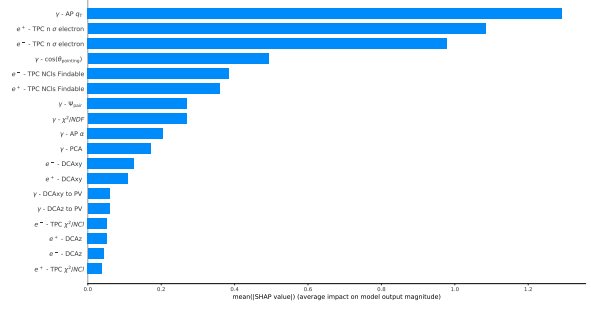
(c) 0.6-0.8 GeV/c



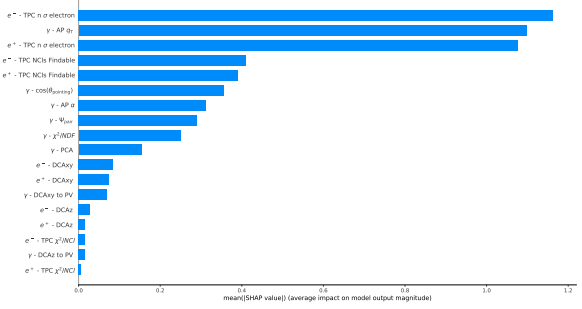
(d) 0.8-1.0 GeV/c



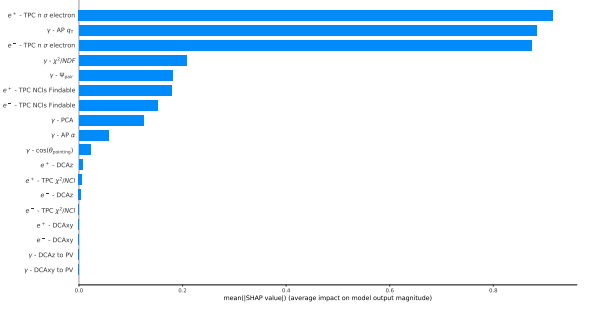
(e) 1.0-1.5 GeV/c



(f) 1.5-2.0 GeV/c

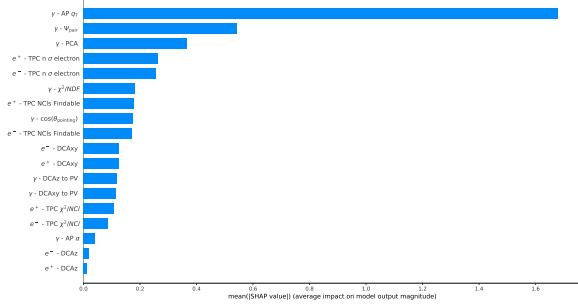


(g) 2.0-5.0 GeV/c

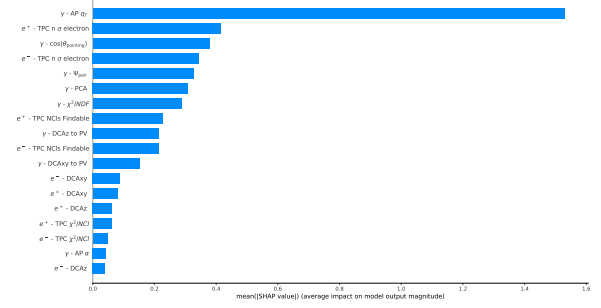


(h) 5.0-10000.0 GeV/c

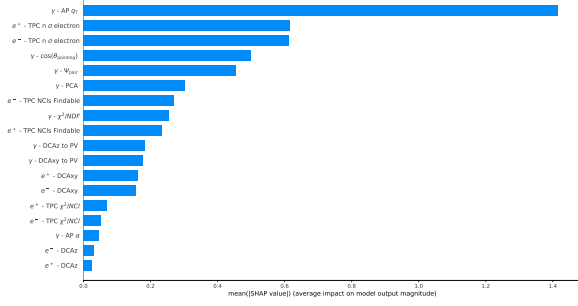
Figure 12: Importance of the input features for all p_T bins in the centrality range 10-20 %



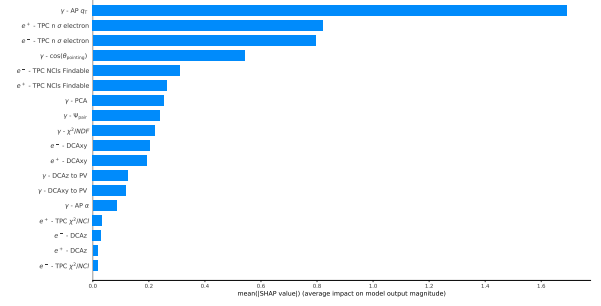
(a) 0.3-0.6 GeV/c



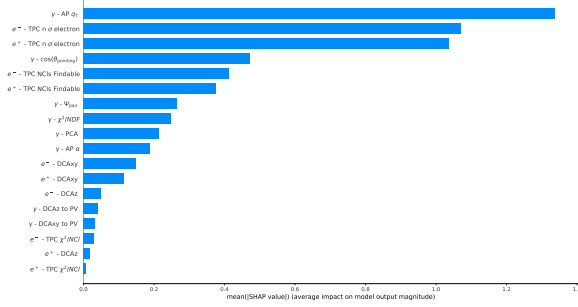
(b) 0.6-0.8 GeV/c



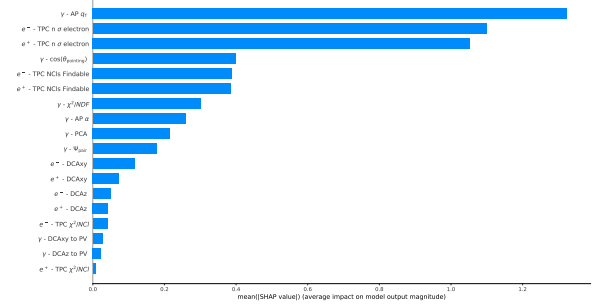
(c) 0.8-1.0 GeV/c



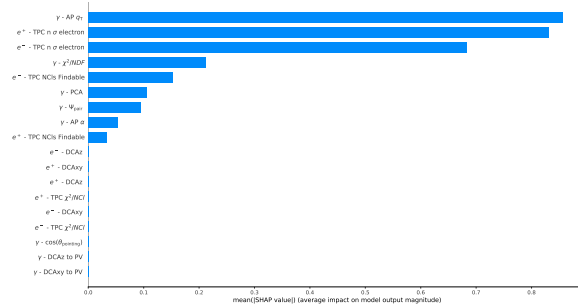
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

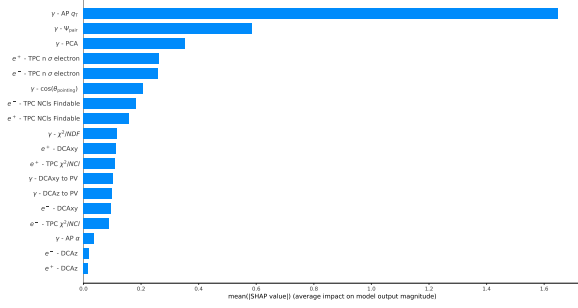


(f) 2.0-5.0 GeV/c

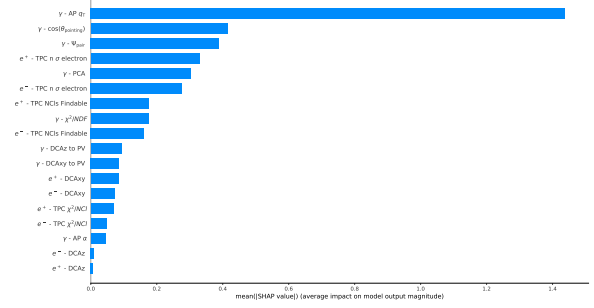


(g) 5.0-10000.0 GeV/c

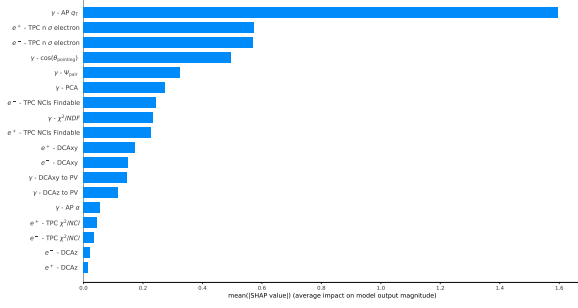
Figure 13: Importance of the input features for all p_T bins in the centrality range 20-30 %



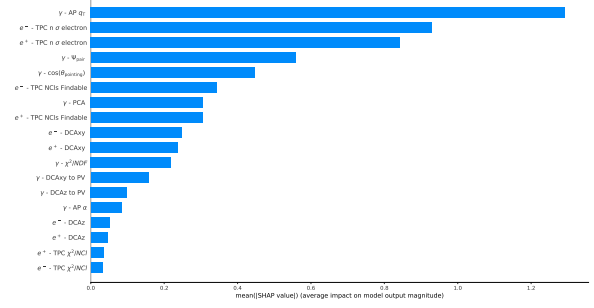
(a) 0.3-0.6 GeV/c



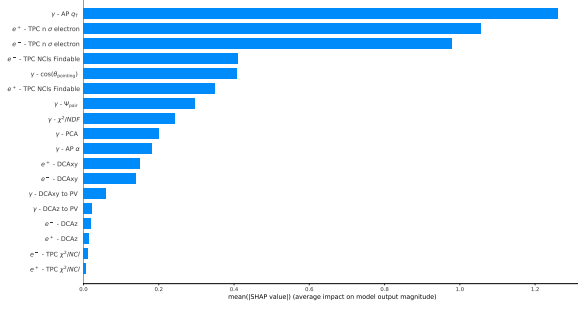
(b) 0.6-0.8 GeV/c



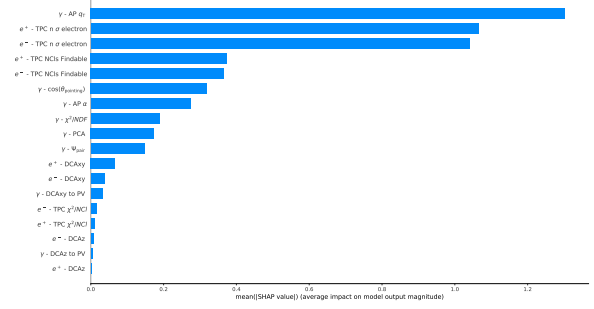
(c) 0.8-1.0 GeV/c



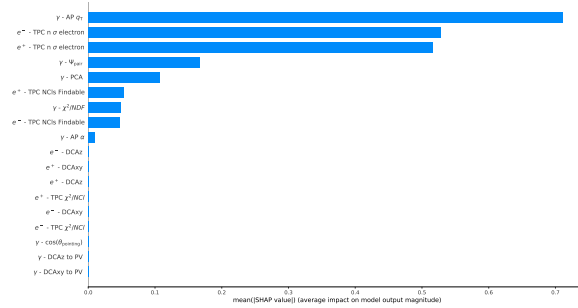
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

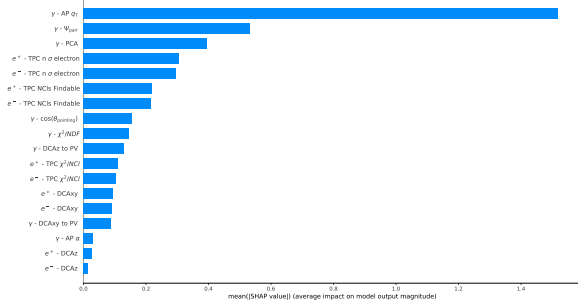


(f) 2.0-5.0 GeV/c

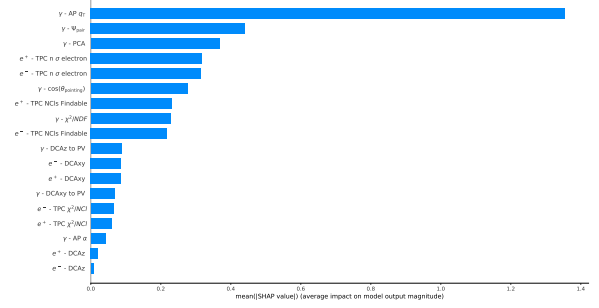


(g) 5.0-10000.0 GeV/c

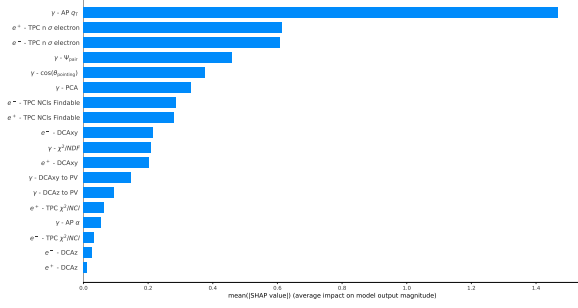
Figure 14: Importance of the input features for all p_T bins in the centrality range 30-40 %



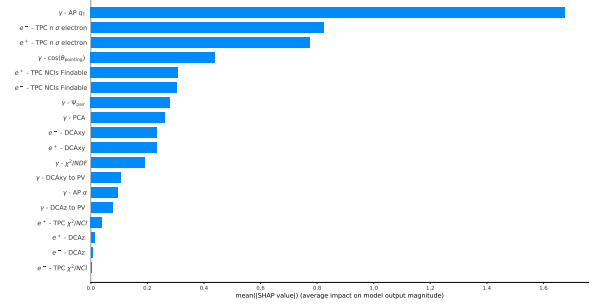
(a) 0.3-0.6 GeV/c



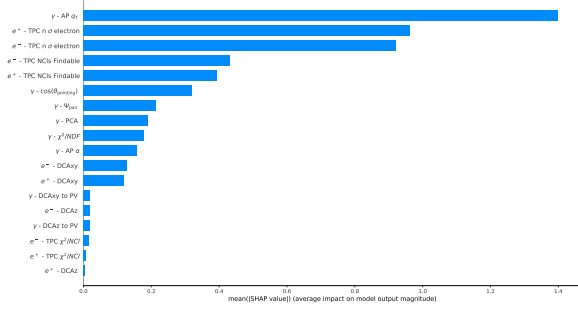
(b) 0.6-0.8 GeV/c



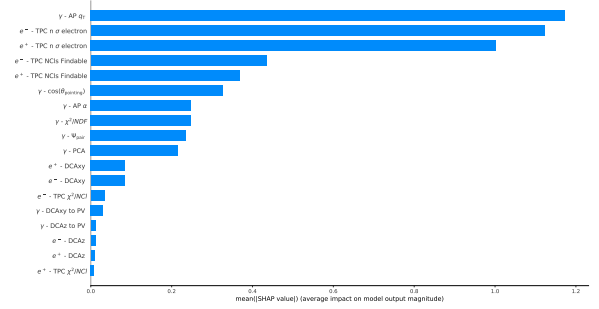
(c) 0.8-1.0 GeV/c



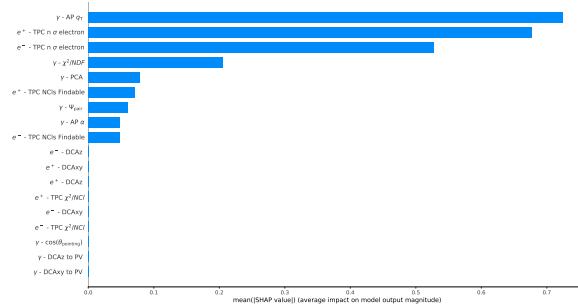
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c



(f) 2.0-5.0 GeV/c



(g) 5.0-10000.0 GeV/c

Figure 15: Importance of the input features for all p_T bins in the centrality range 40-50 %

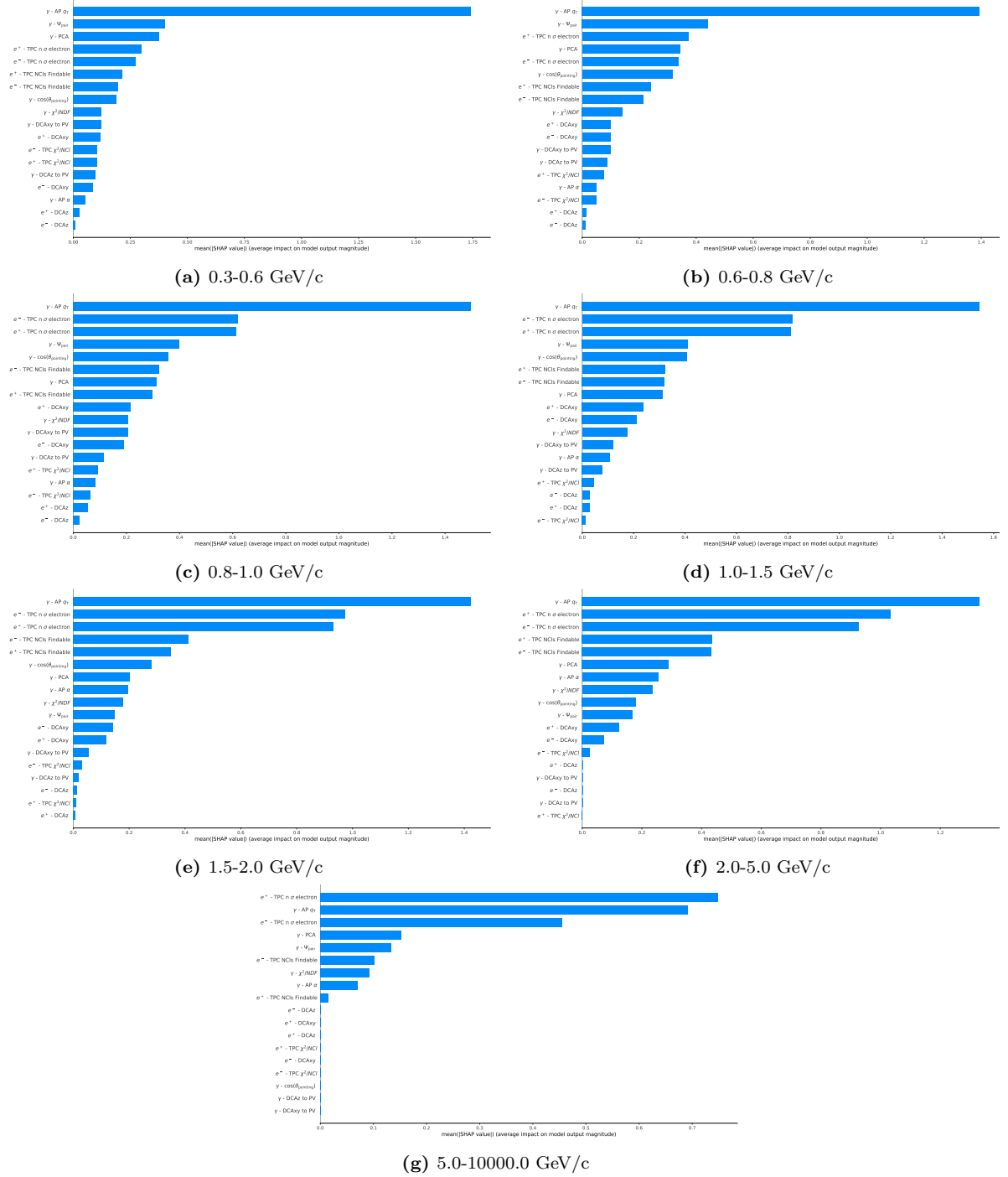
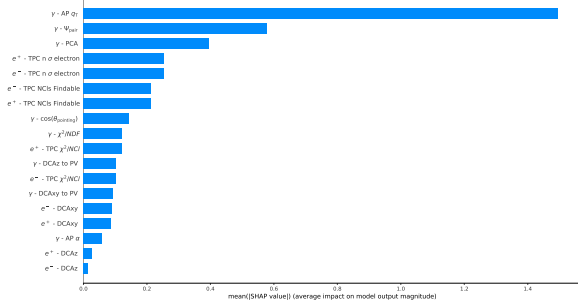
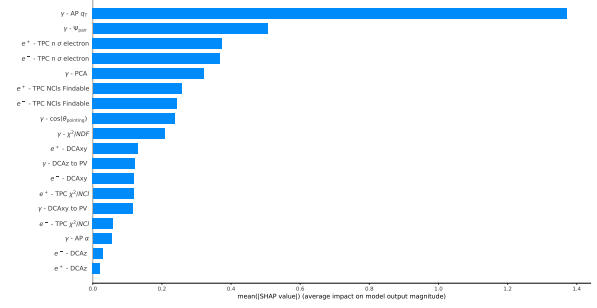


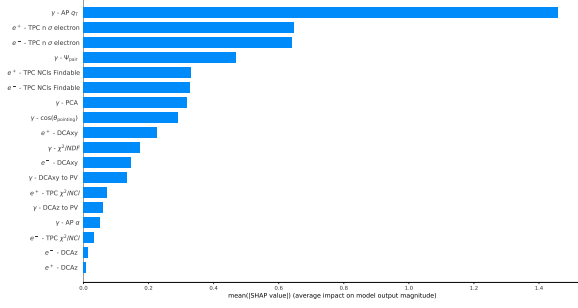
Figure 16: Importance of the input features for all p_T bins in the centrality range 50-60 %



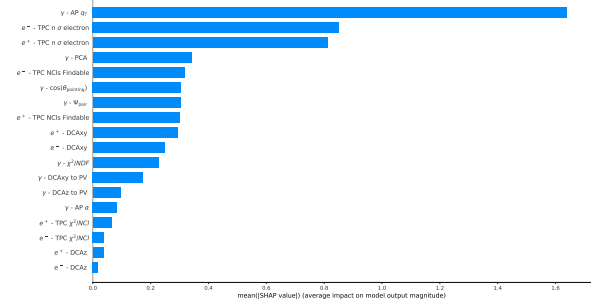
(a) 0.3-0.6 GeV/c



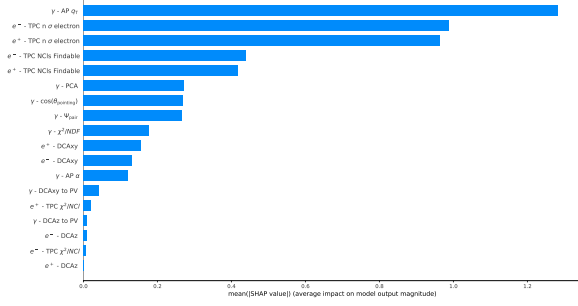
(b) 0.6-0.8 GeV/c



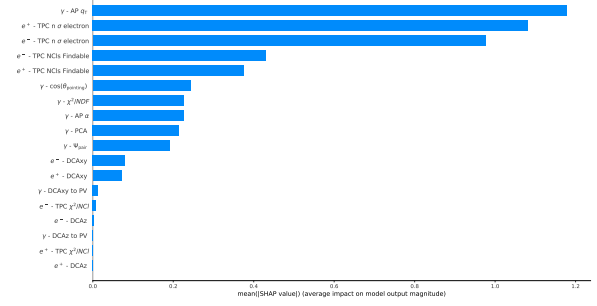
(c) 0.8-1.0 GeV/c



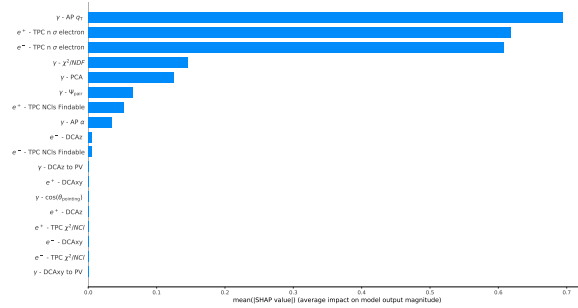
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

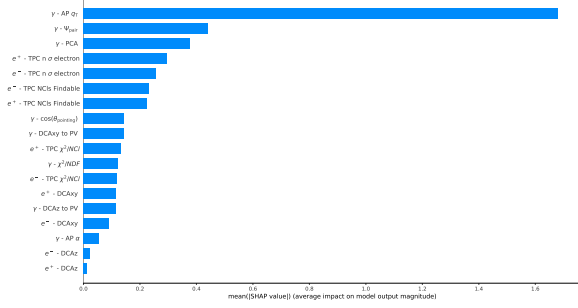


(f) 2.0-5.0 GeV/c

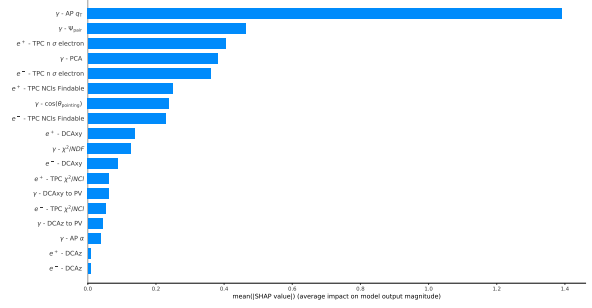


(g) 5.0-10000.0 GeV/c

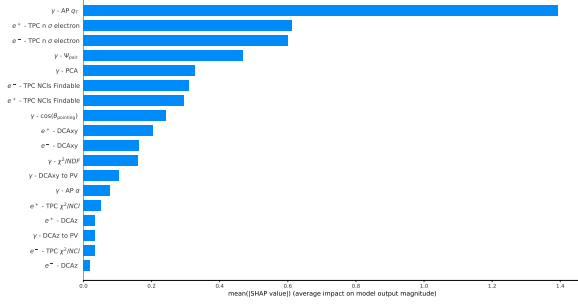
Figure 17: Importance of the input features for all p_T bins in the centrality range 60-70 %



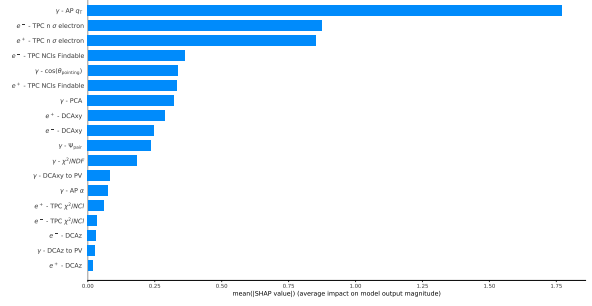
(a) 0.3-0.6 GeV/c



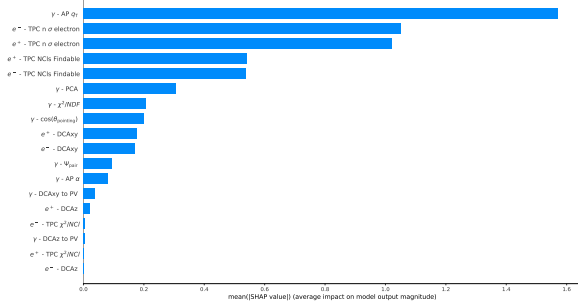
(b) 0.6-0.8 GeV/c



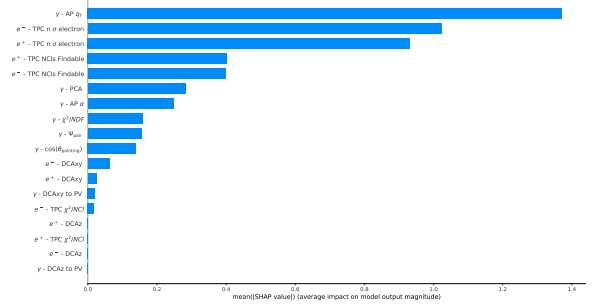
(c) 0.8-1.0 GeV/c



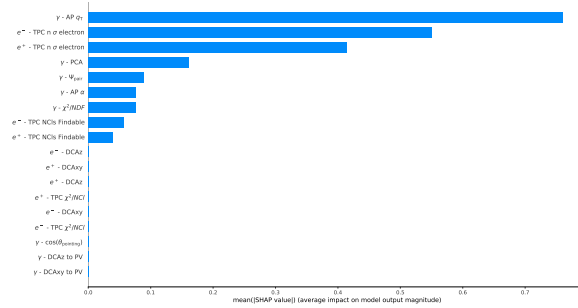
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

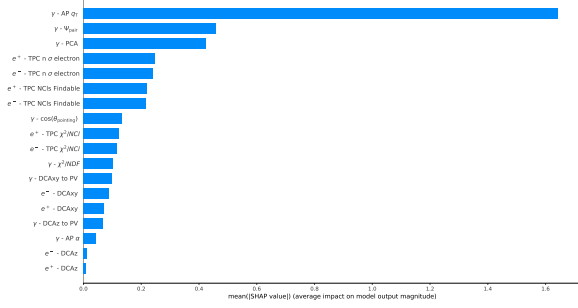


(f) 2.0-5.0 GeV/c

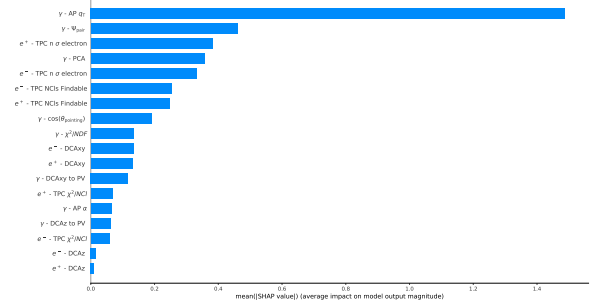


(g) 5.0-10000.0 GeV/c

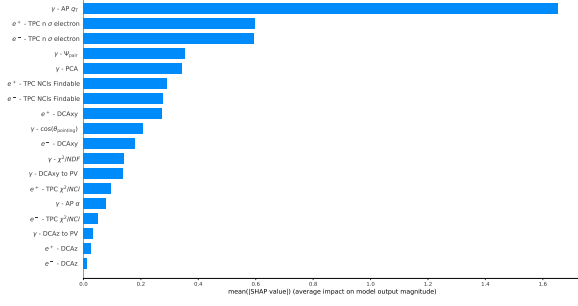
Figure 18: Importance of the input features for all p_T bins in the centrality range 70-80 %



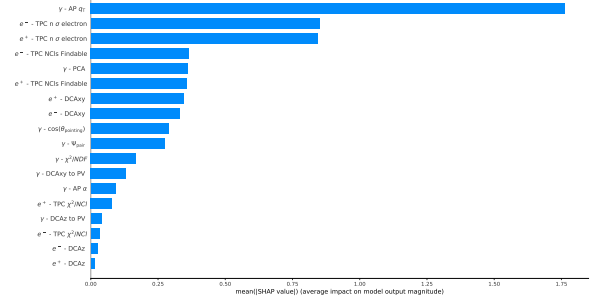
(a) 0.3-0.6 GeV/c



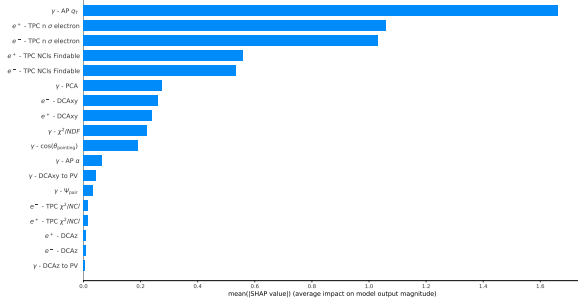
(b) 0.6-0.8 GeV/c



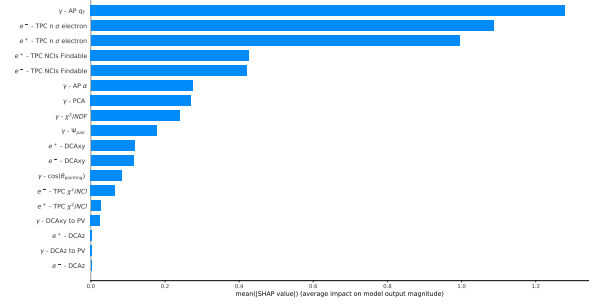
(c) 0.8-1.0 GeV/c



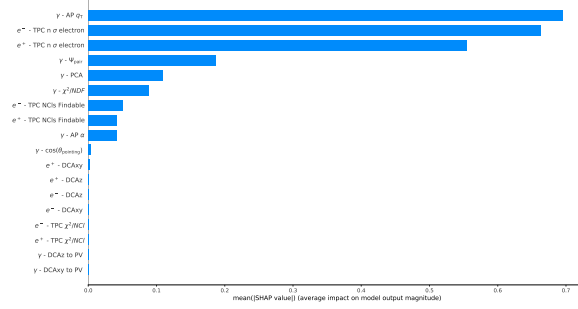
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c



(f) 2.0-5.0 GeV/c



(g) 5.0-10000.0 GeV/c

Figure 20: Importance of the input features for all p_T bins in the centrality range 90-100 %

3 Correlation map of the input features

3.1 Correlations of primary photons

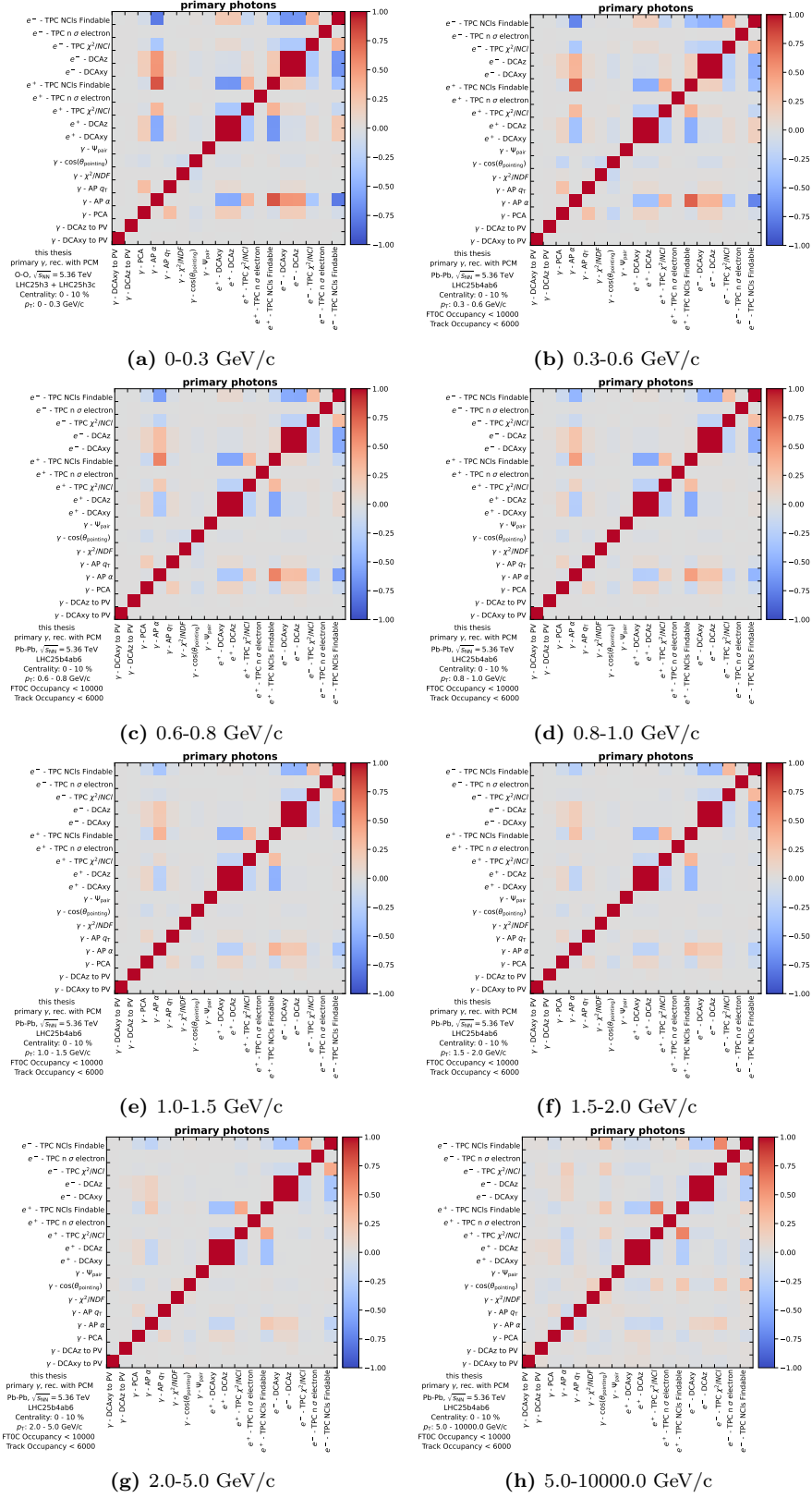


Figure 21: Correlation map of primary photons for all p_T bins in the centrality range 0-10 %

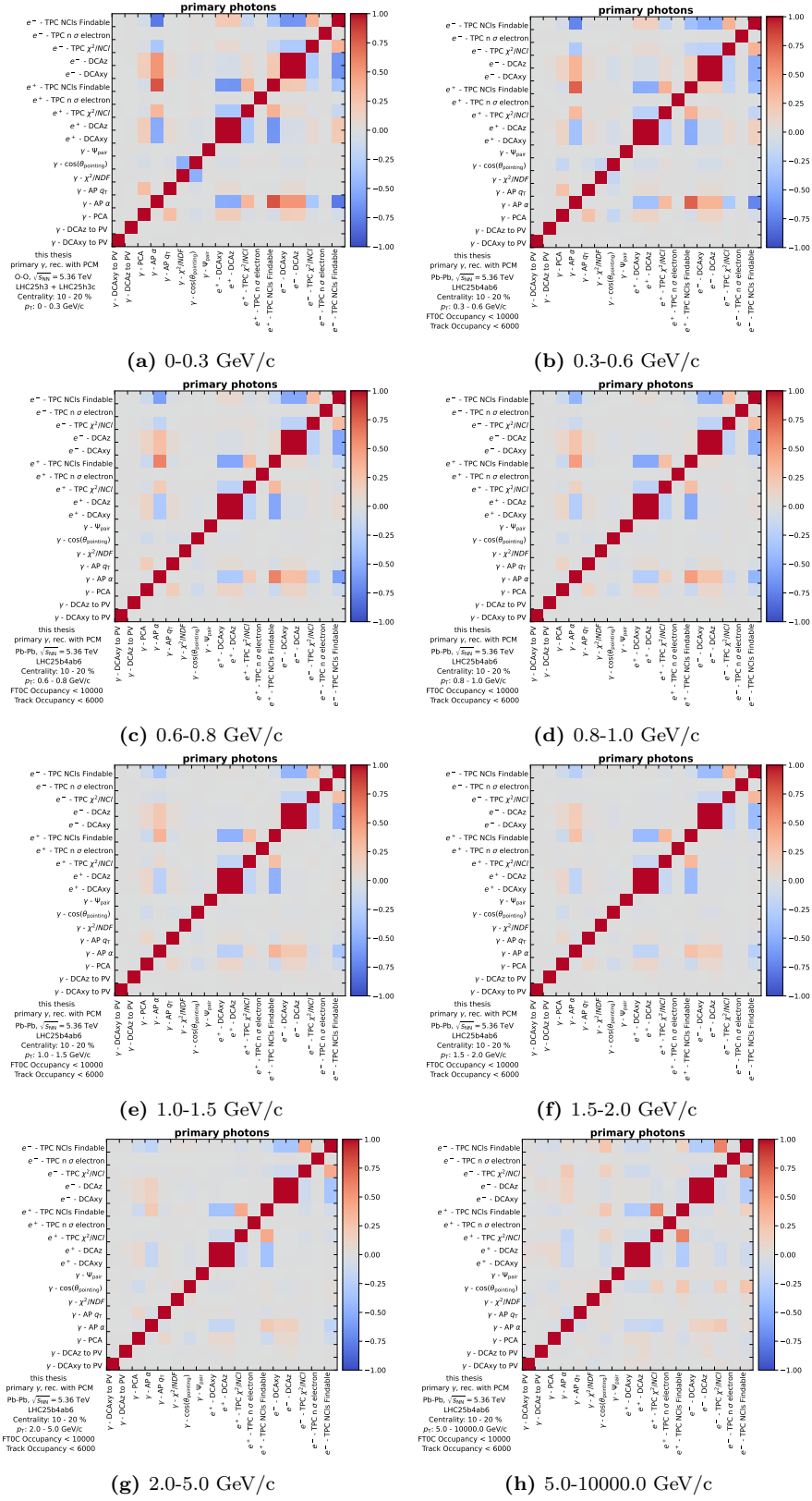


Figure 22: Correlation map of primary photons for all p_T bins in the centrality range 10-20 %

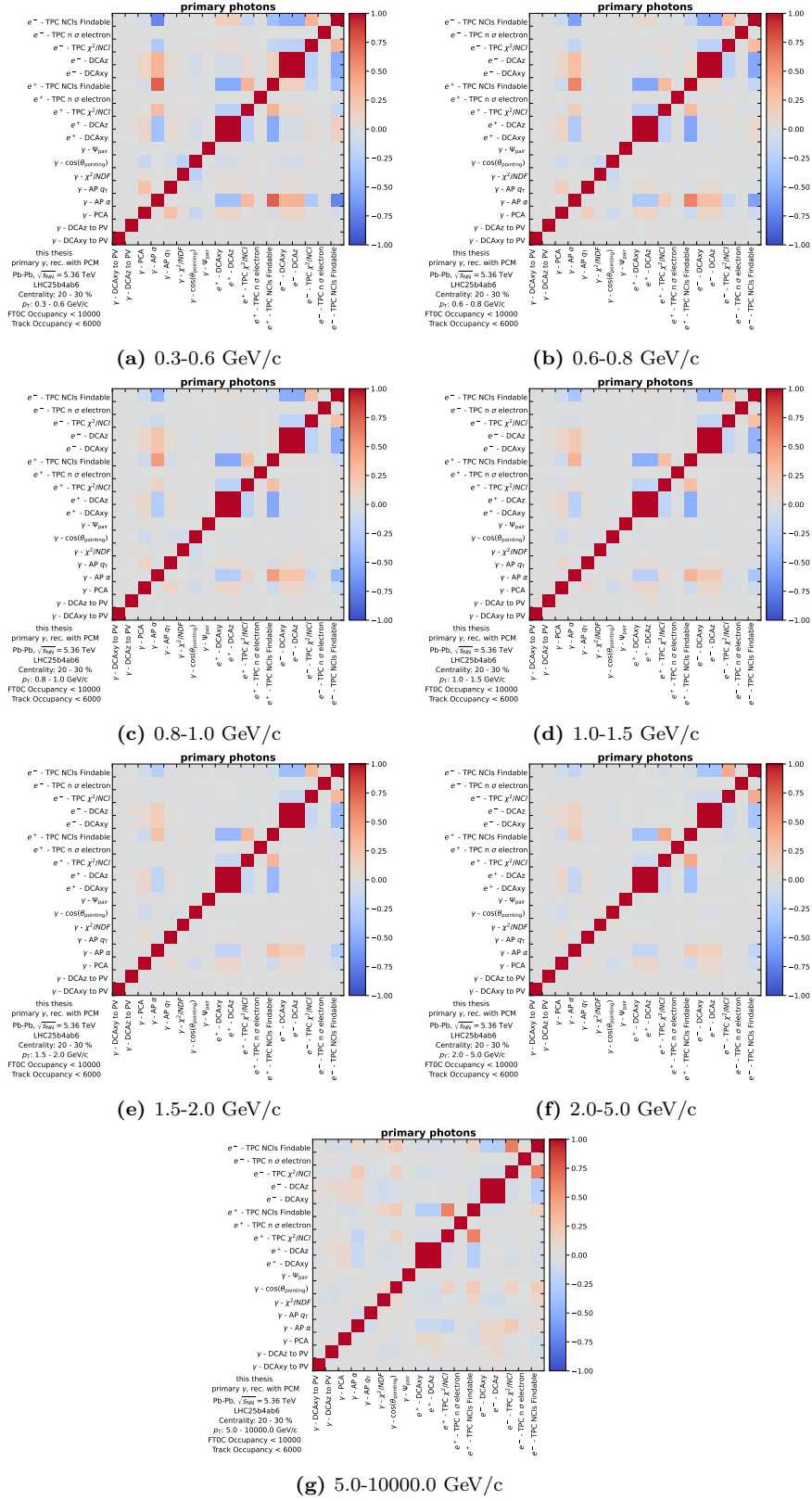


Figure 23: Correlation map of primary photons for all p_T bins in the centrality range 20-30 %

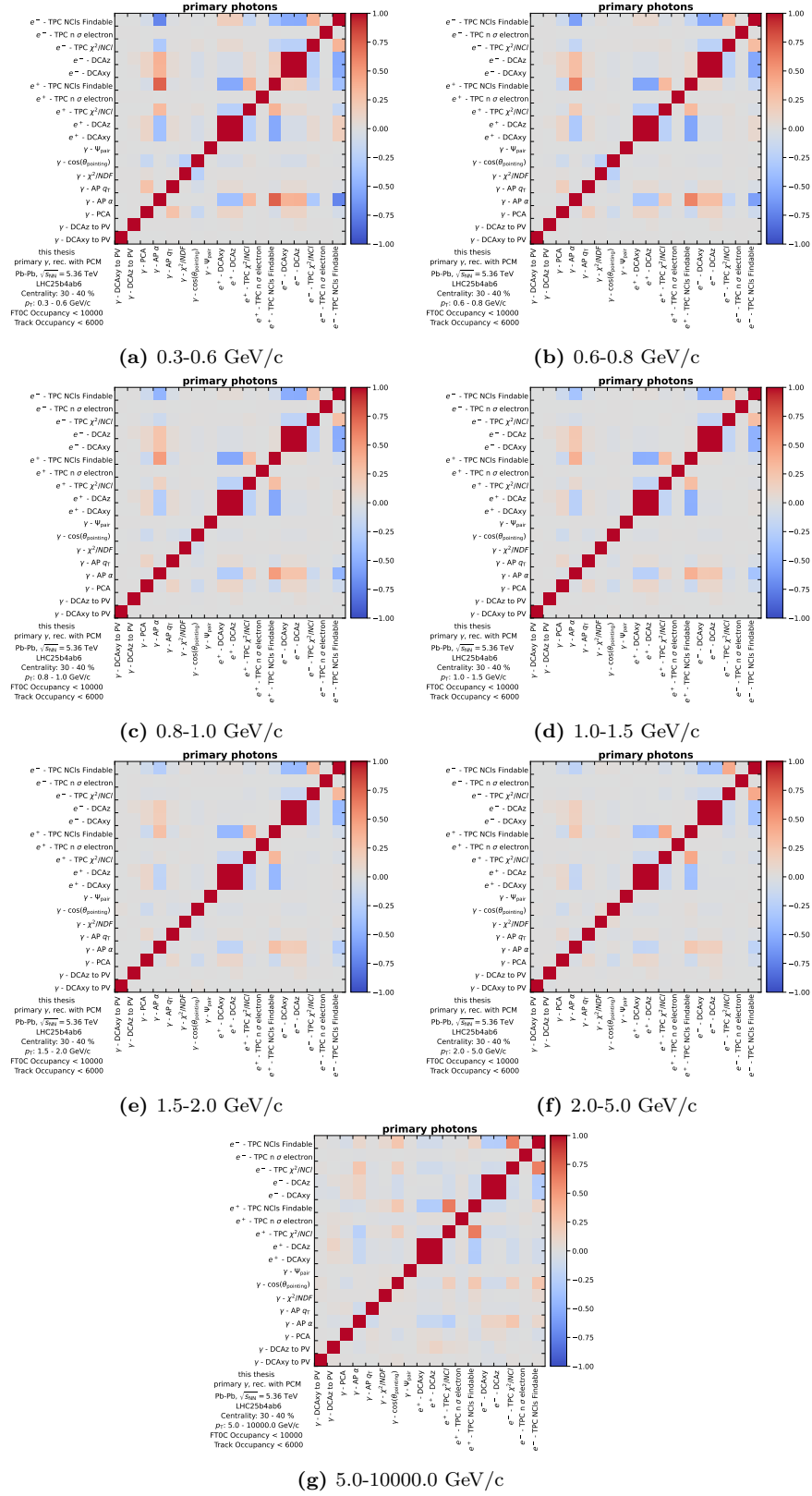


Figure 24: Correlation map of primary photons for all p_T bins in the centrality range 30-40 %

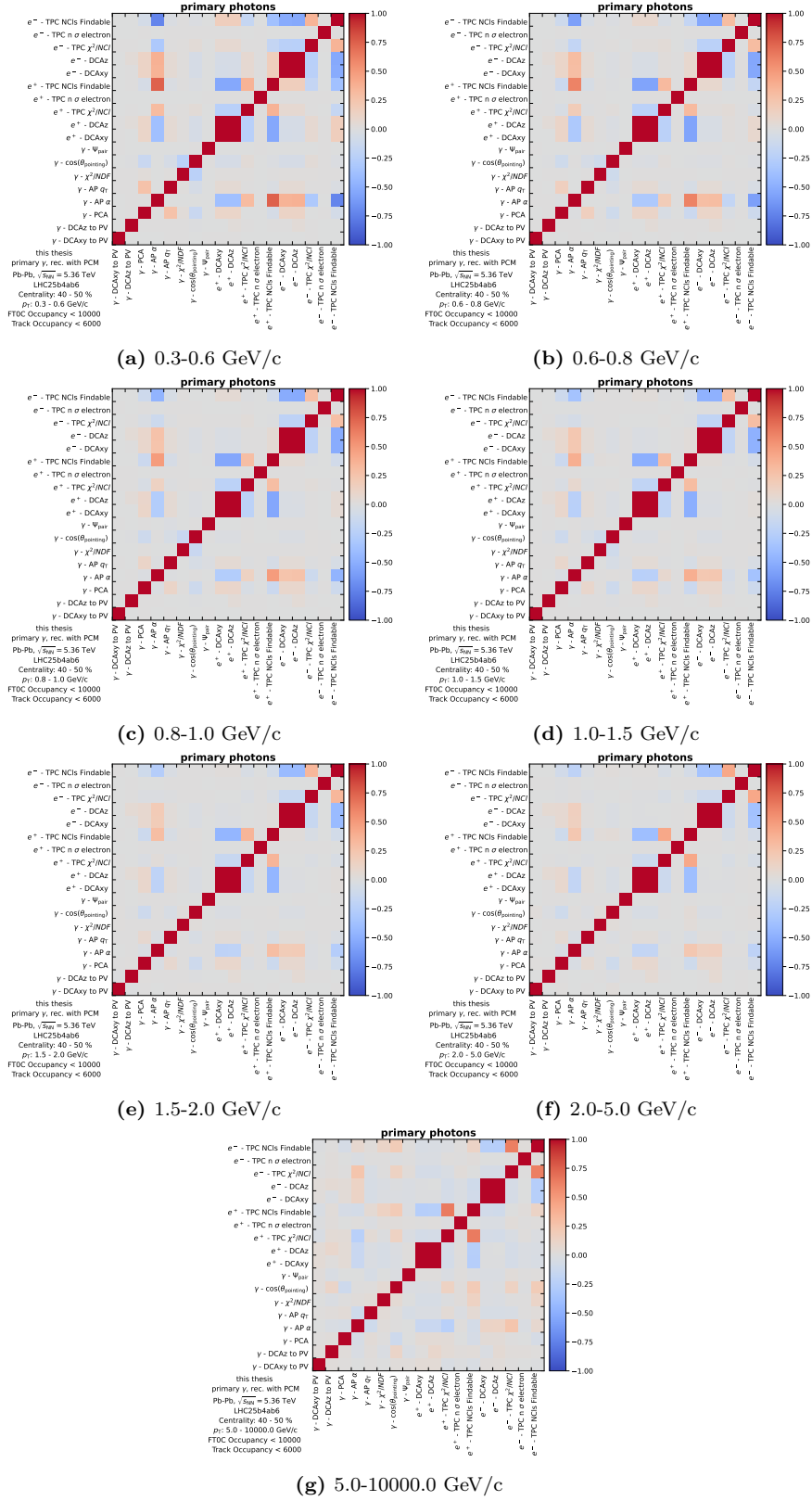


Figure 25: Correlation map of primary photons for all p_T bins in the centrality range 40-50 %

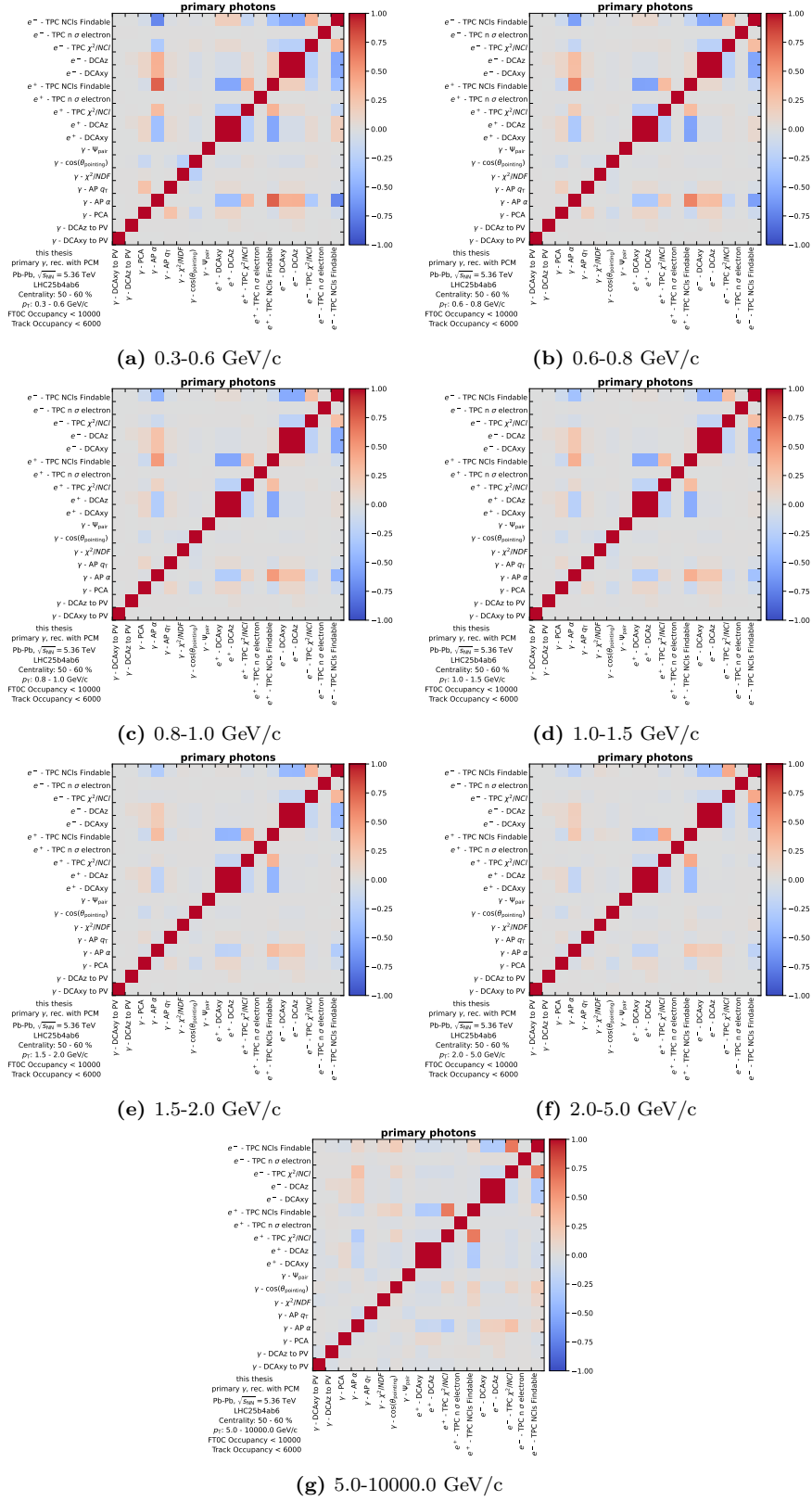


Figure 26: Correlation map of primary photons for all p_T bins in the centrality range 50-60 %

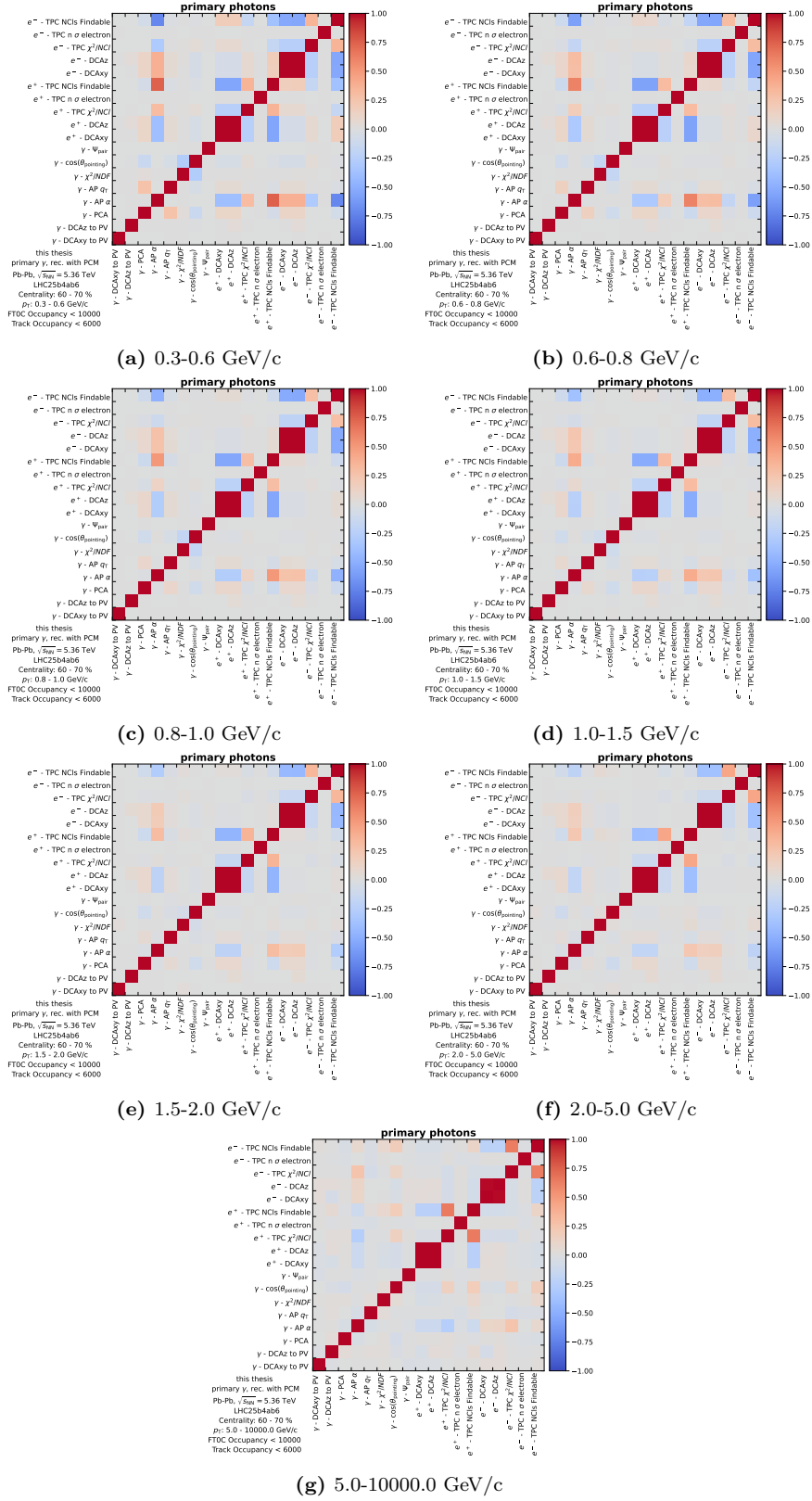


Figure 27: Correlation map of primary photons for all p_T bins in the centrality range 60-70 %

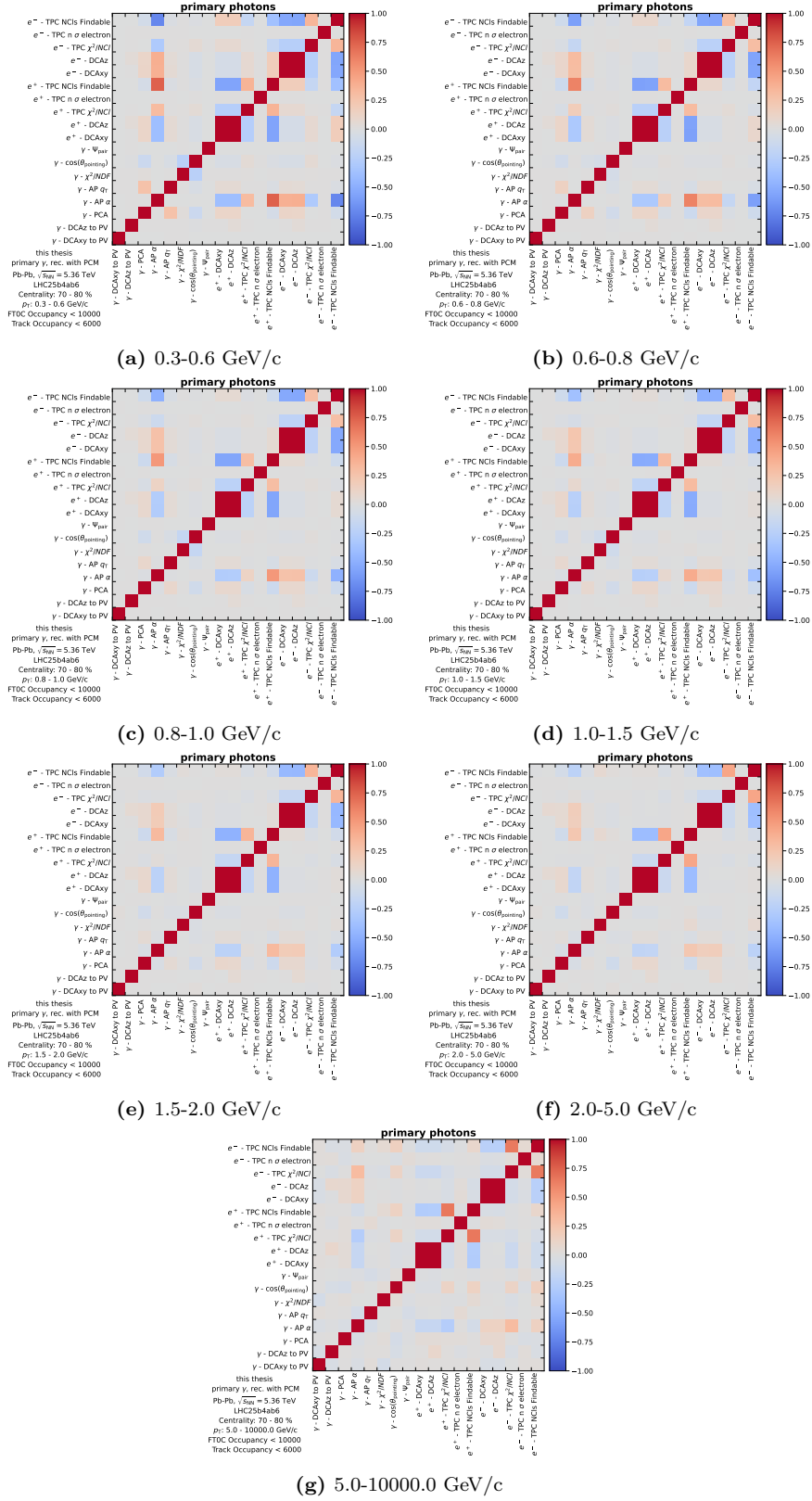


Figure 28: Correlation map of primary photons for all p_T bins in the centrality range 70-80 %

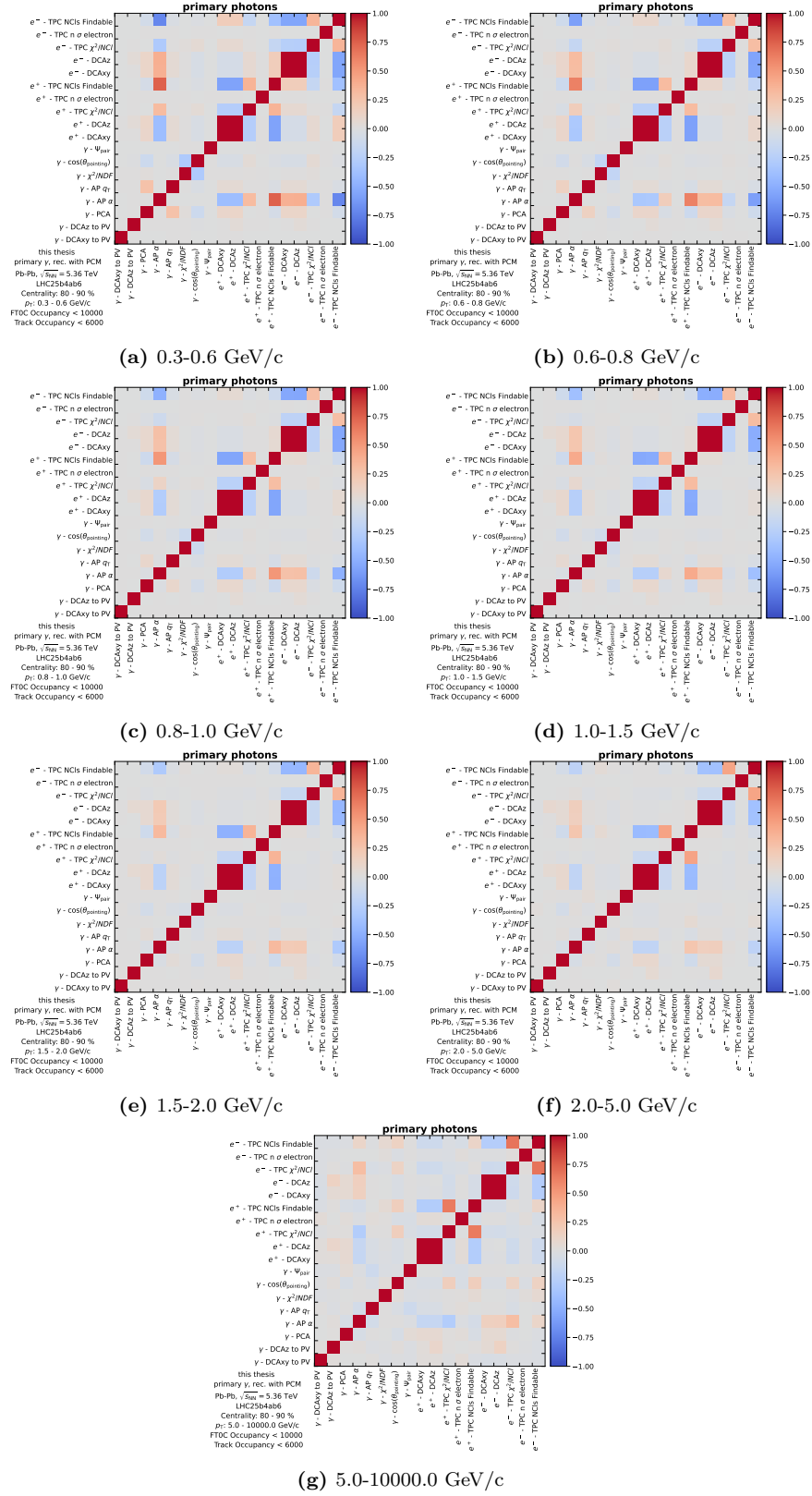


Figure 29: Correlation map of primary photons for all p_T bins in the centrality range 80-90 %

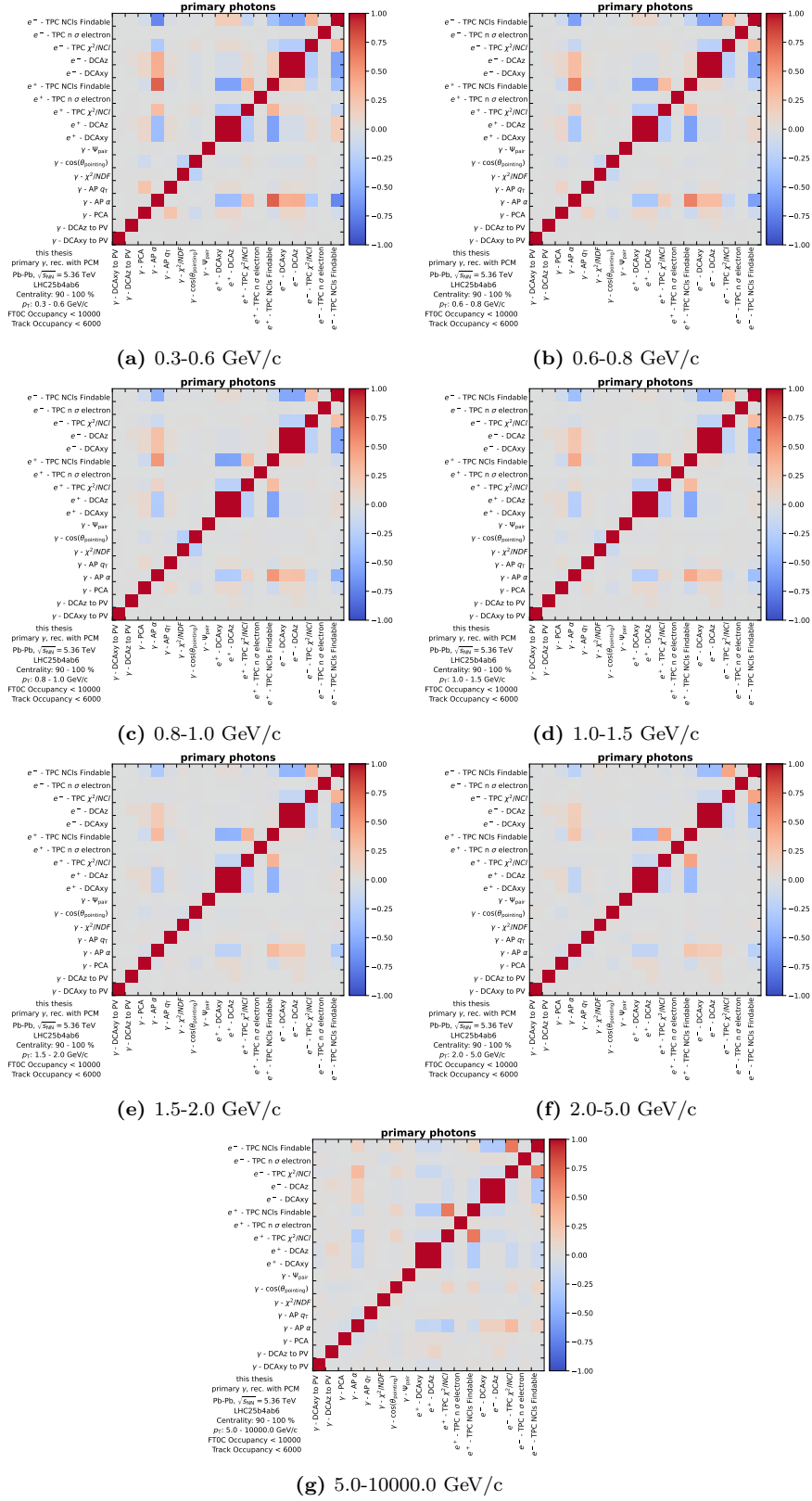


Figure 30: Correlation map of primary photons for all p_T bins in the centrality range 90-100 %

3.2 Correlations of background candidates

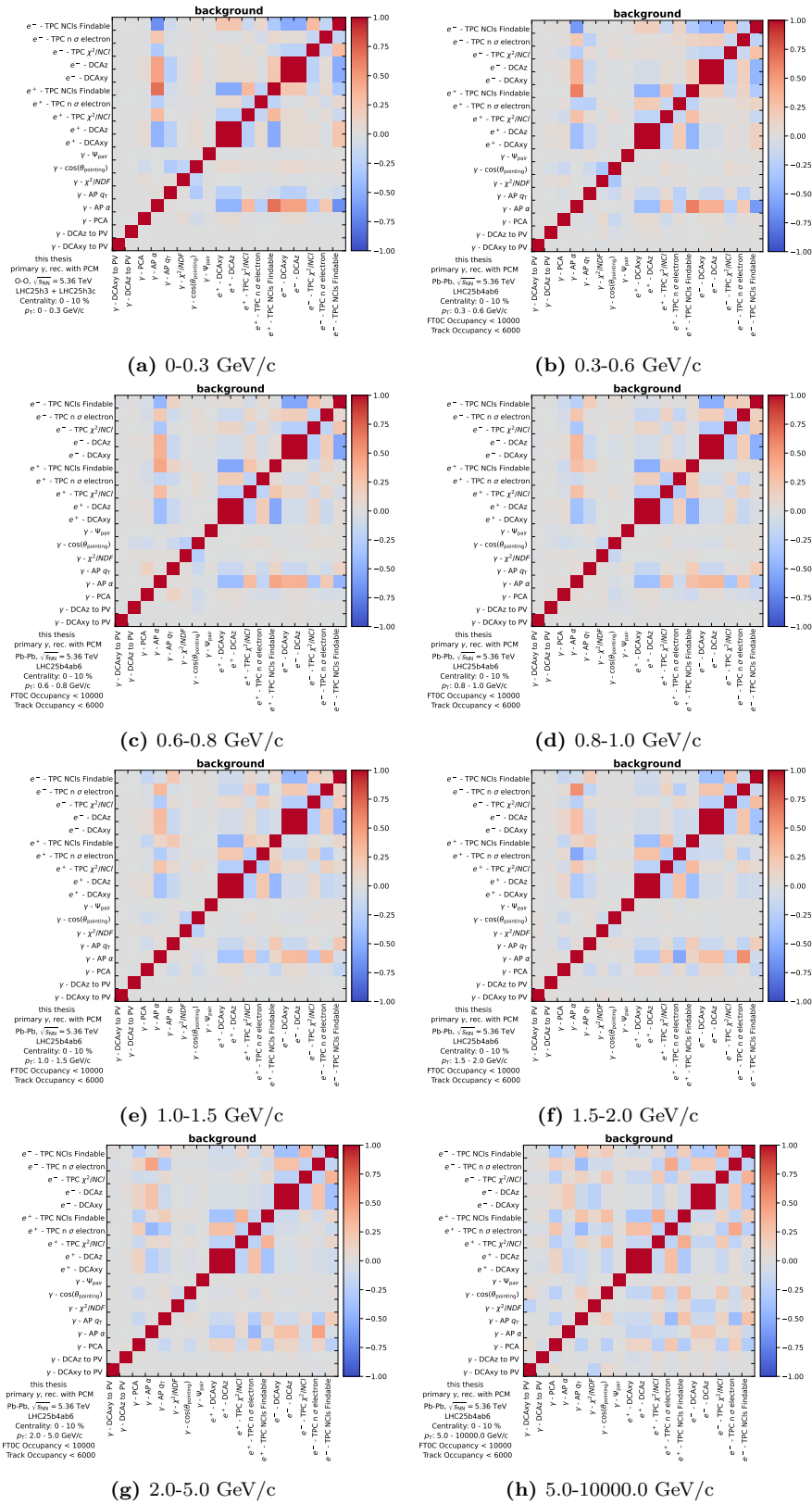


Figure 31: Correlation map of background candidates for all p_T bins in the centrality range 0-10 %

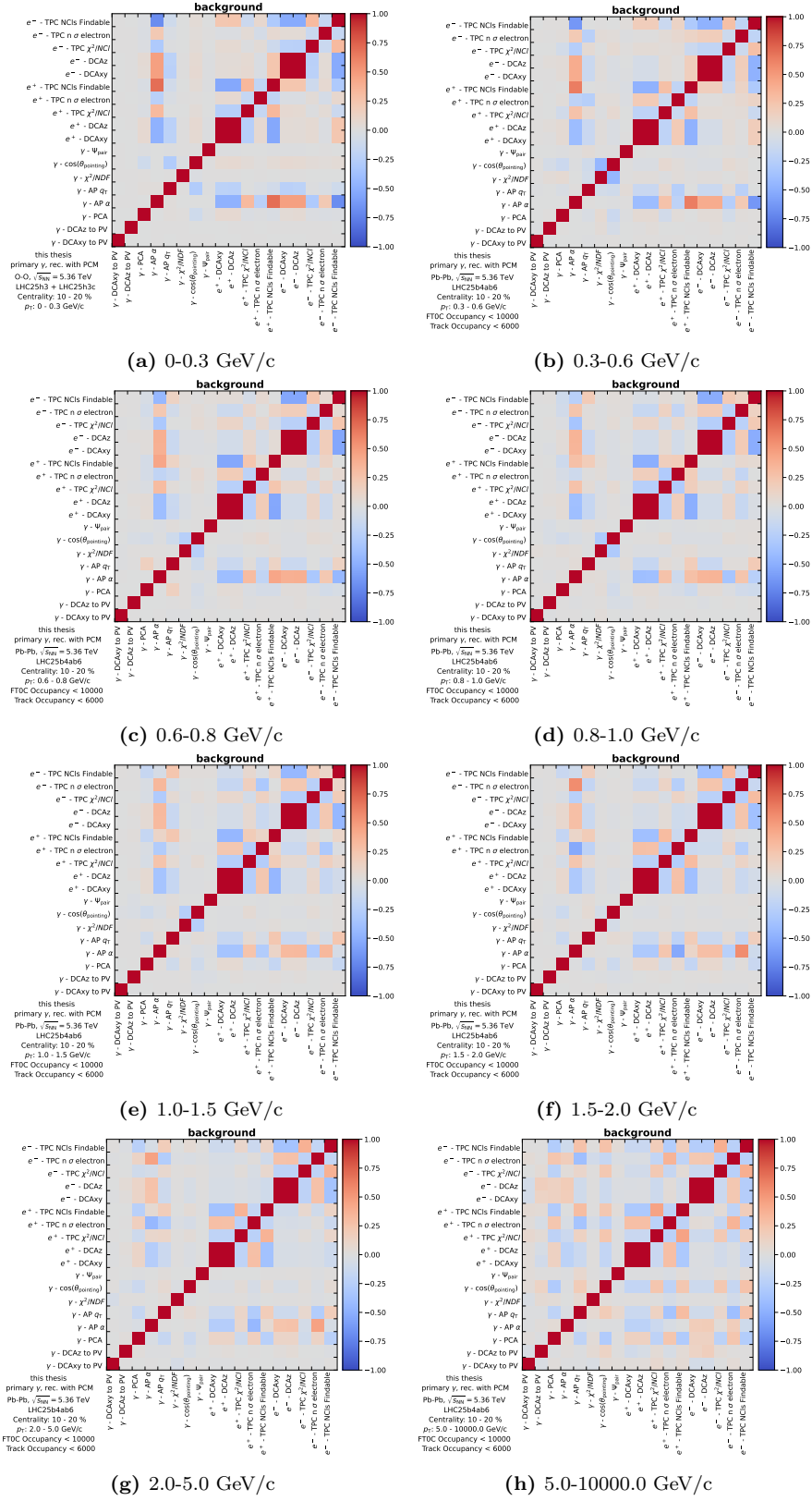


Figure 32: Correlation map of background candidates for all p_T bins in the centrality range 10-20 %

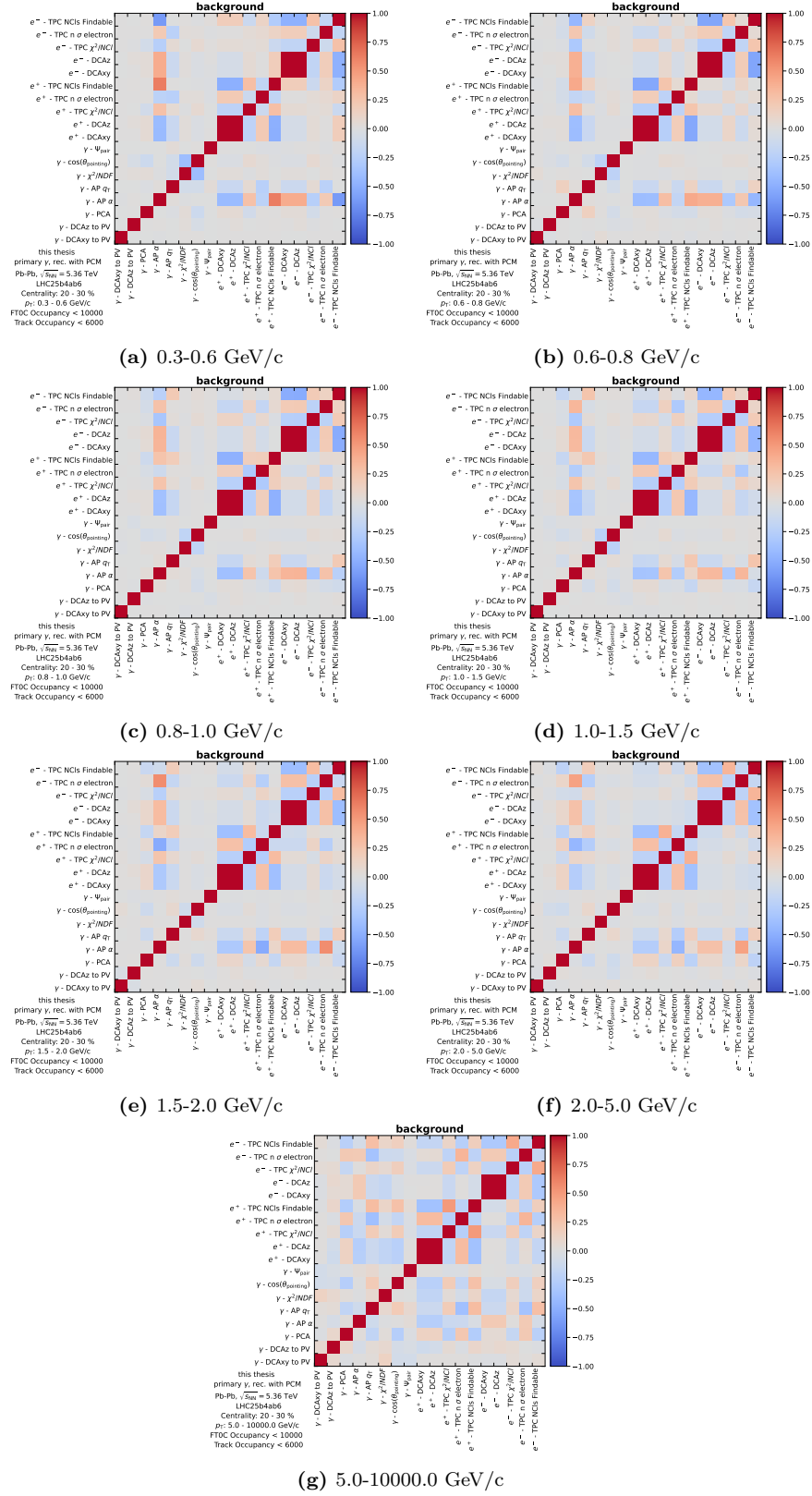


Figure 33: Correlation map of background candidates for all p_T bins in the centrality range 20-30 %

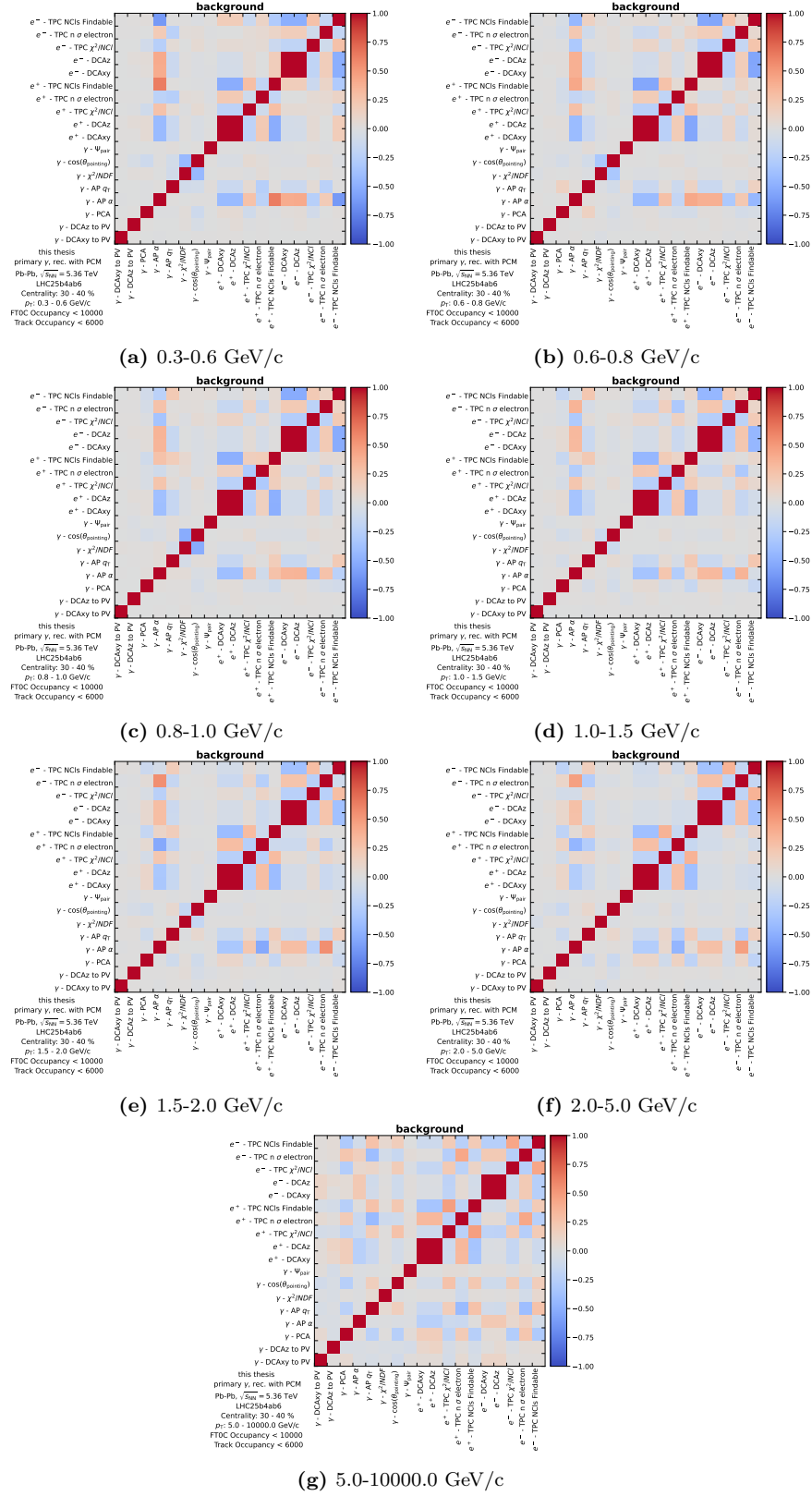


Figure 34: Correlation map of background candidates for all p_T bins in the centrality range 30-40 %

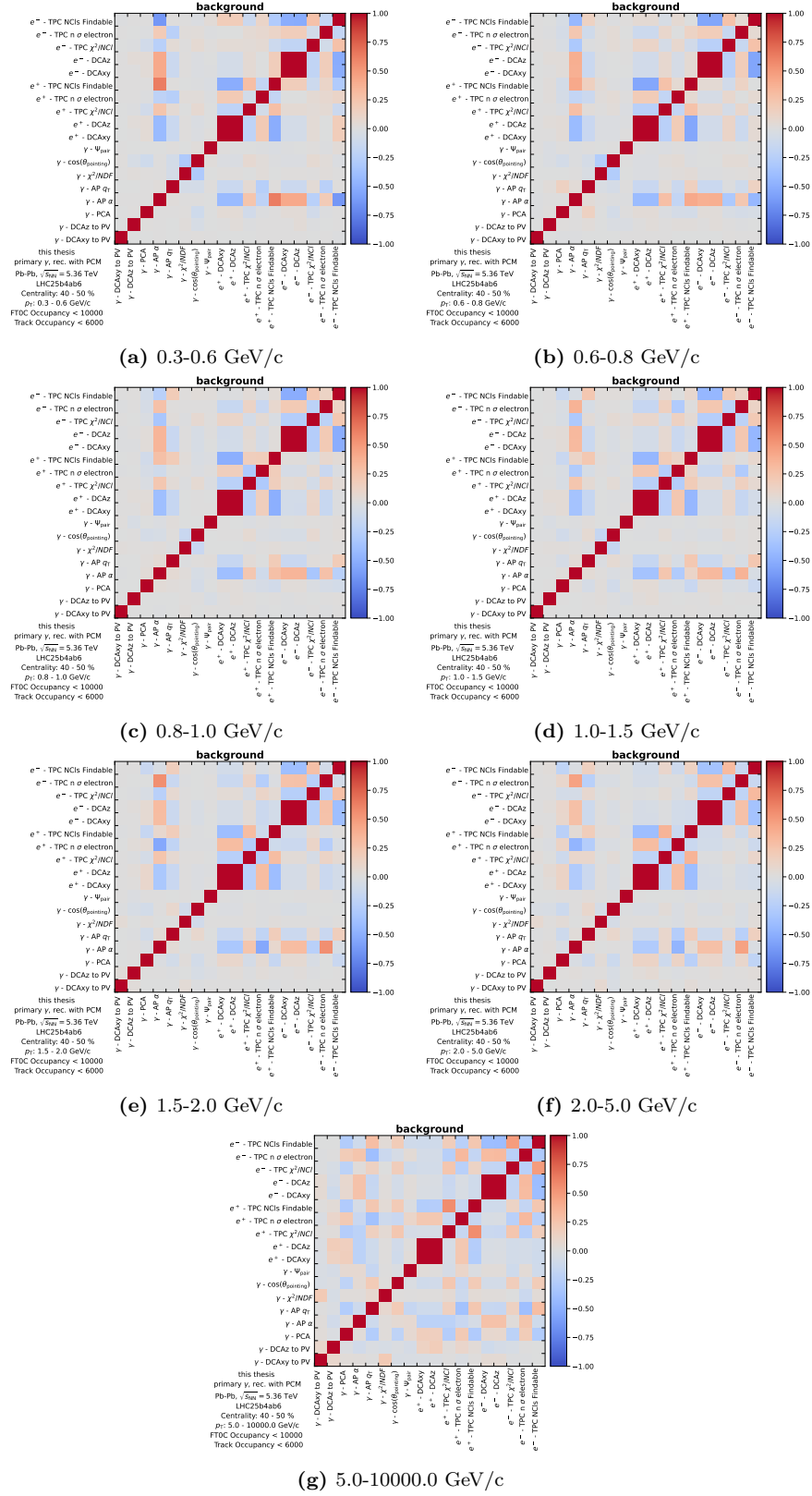


Figure 35: Correlation map of background candidates for all p_T bins in the centrality range 40-50 %

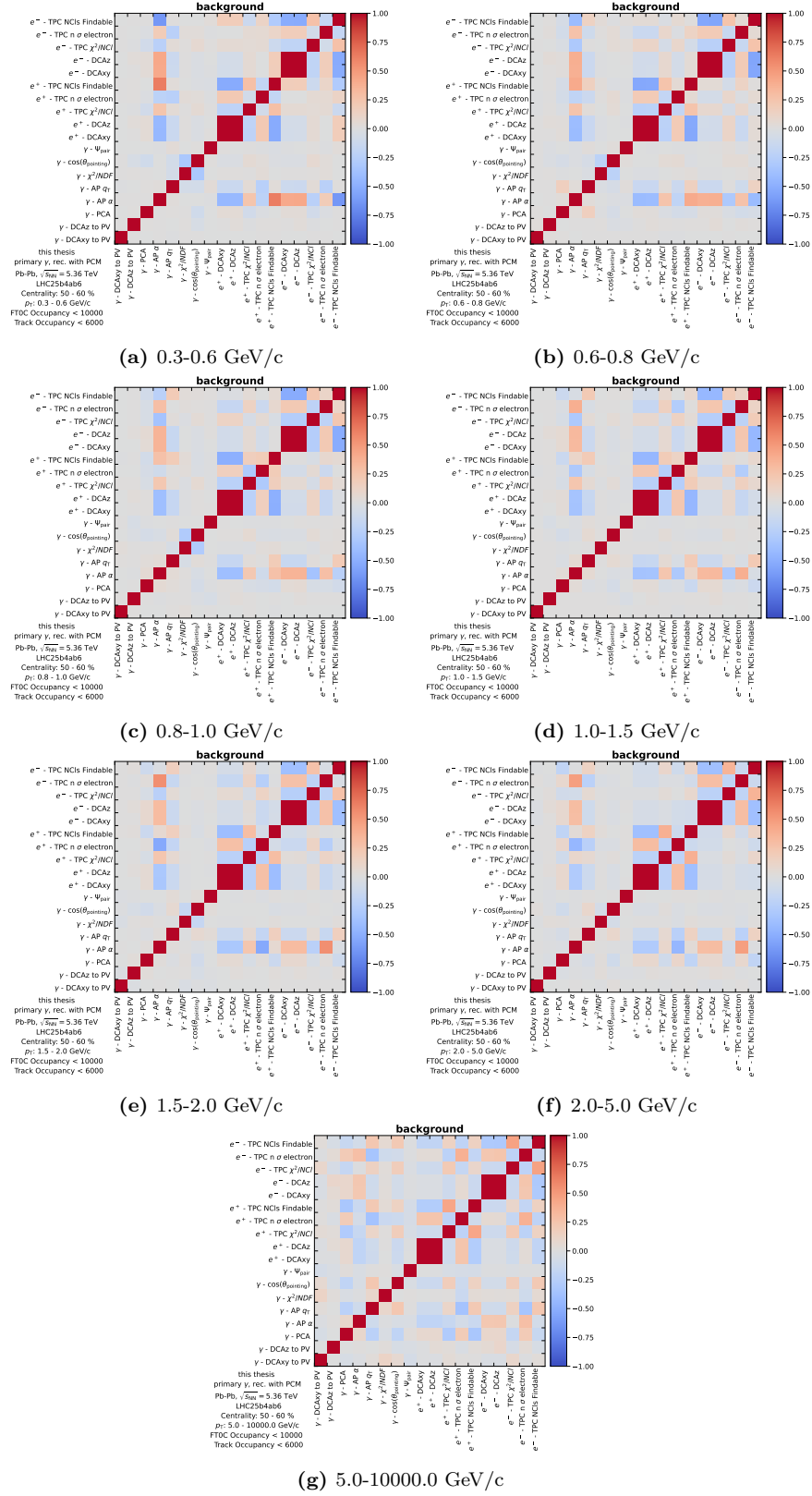


Figure 36: Correlation map of background candidates for all p_T bins in the centrality range 50-60 %

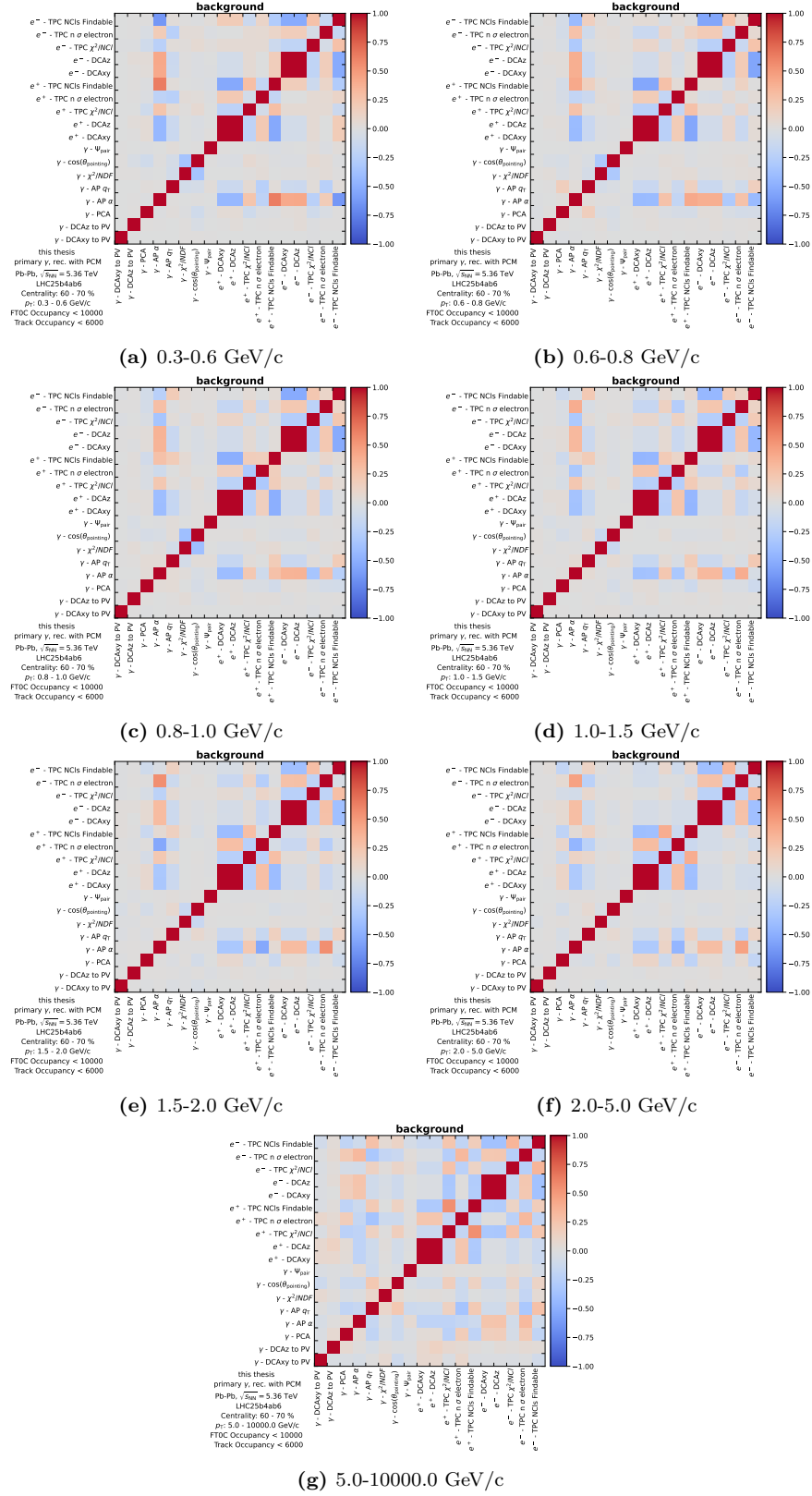


Figure 37: Correlation map of background candidates for all p_T bins in the centrality range 60-70 %

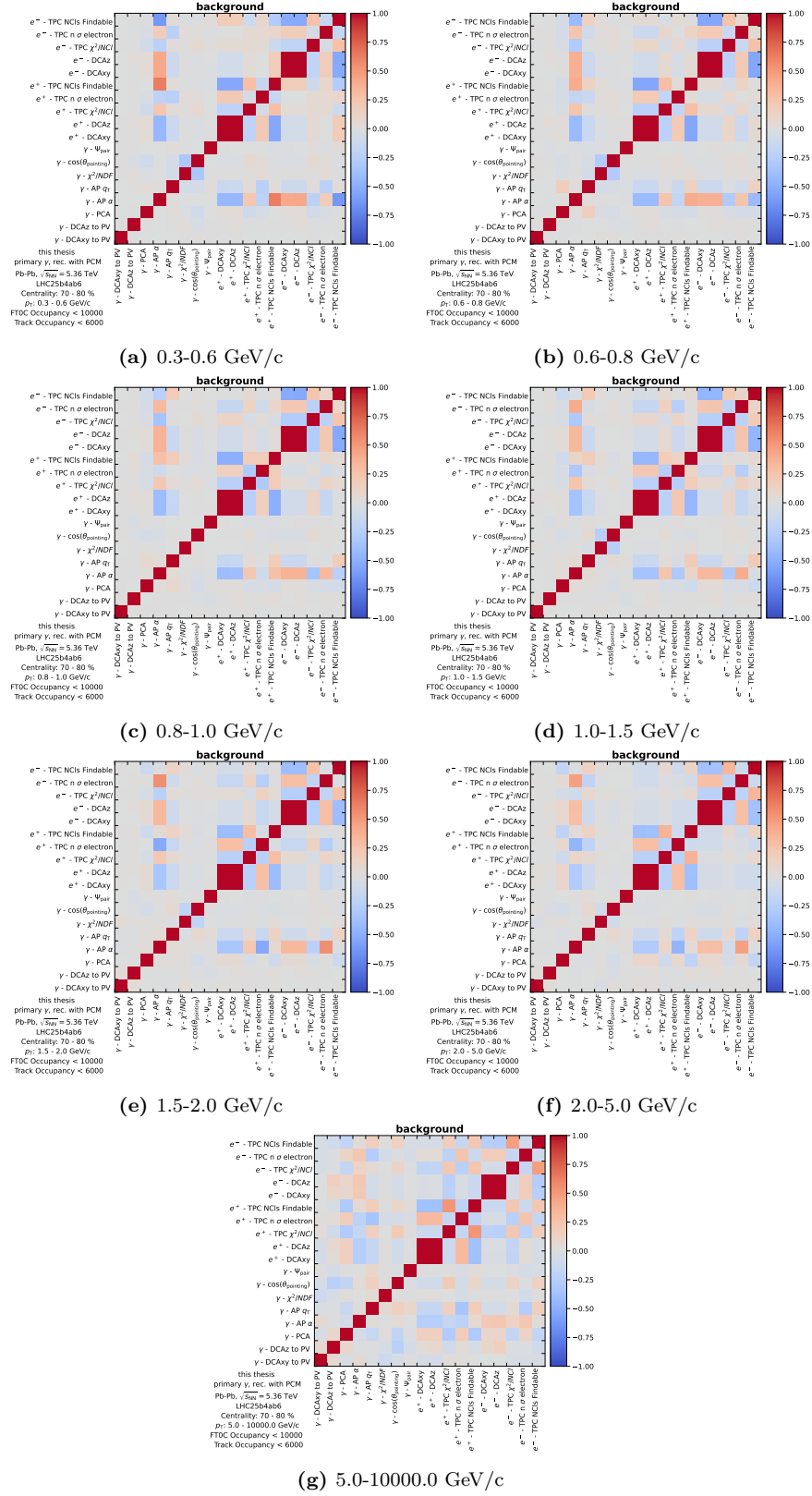


Figure 38: Correlation map of background candidates for all p_T bins in the centrality range 70-80 %

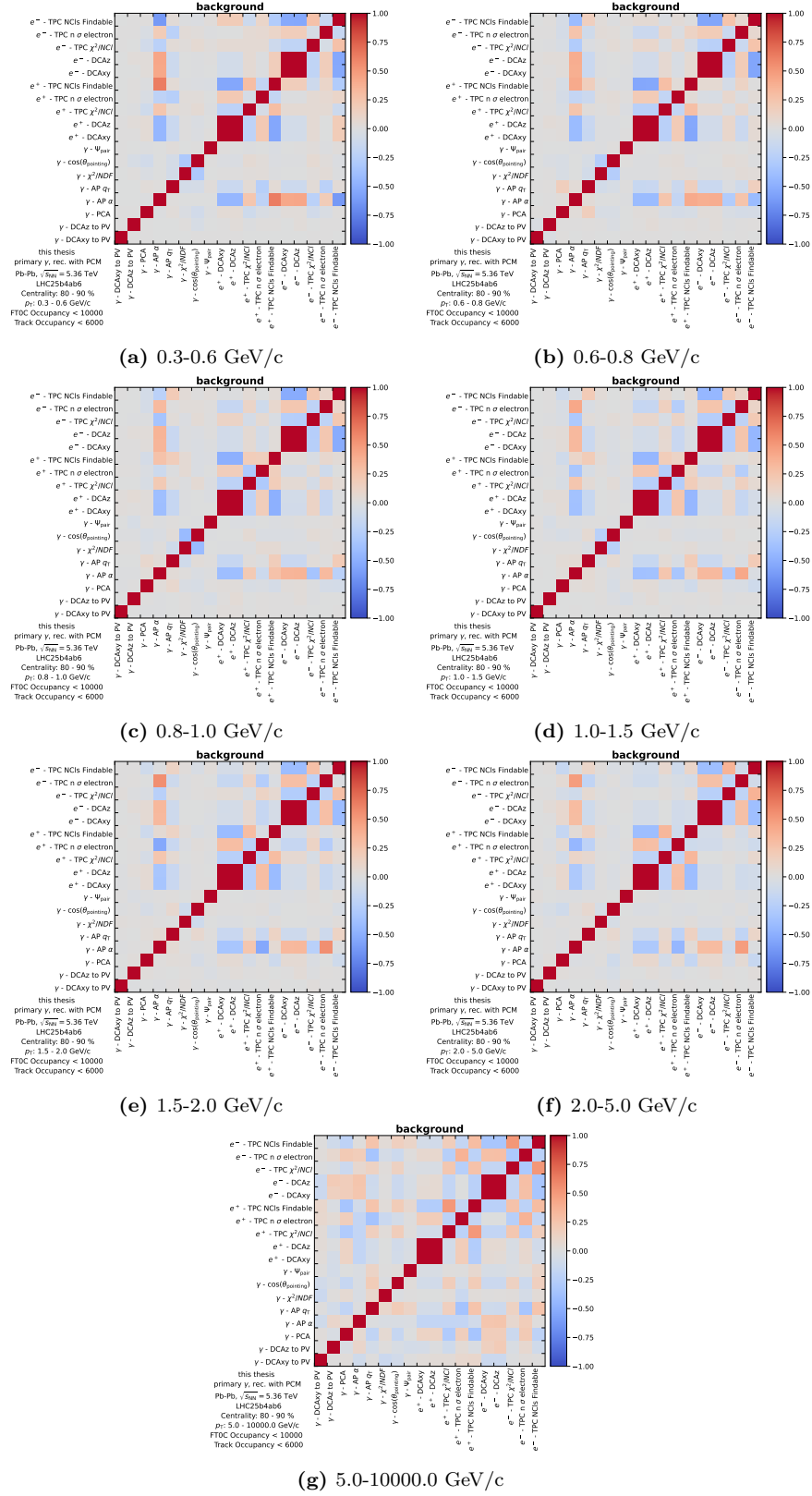


Figure 39: Correlation map of background candidates for all p_T bins in the centrality range 80-90 %

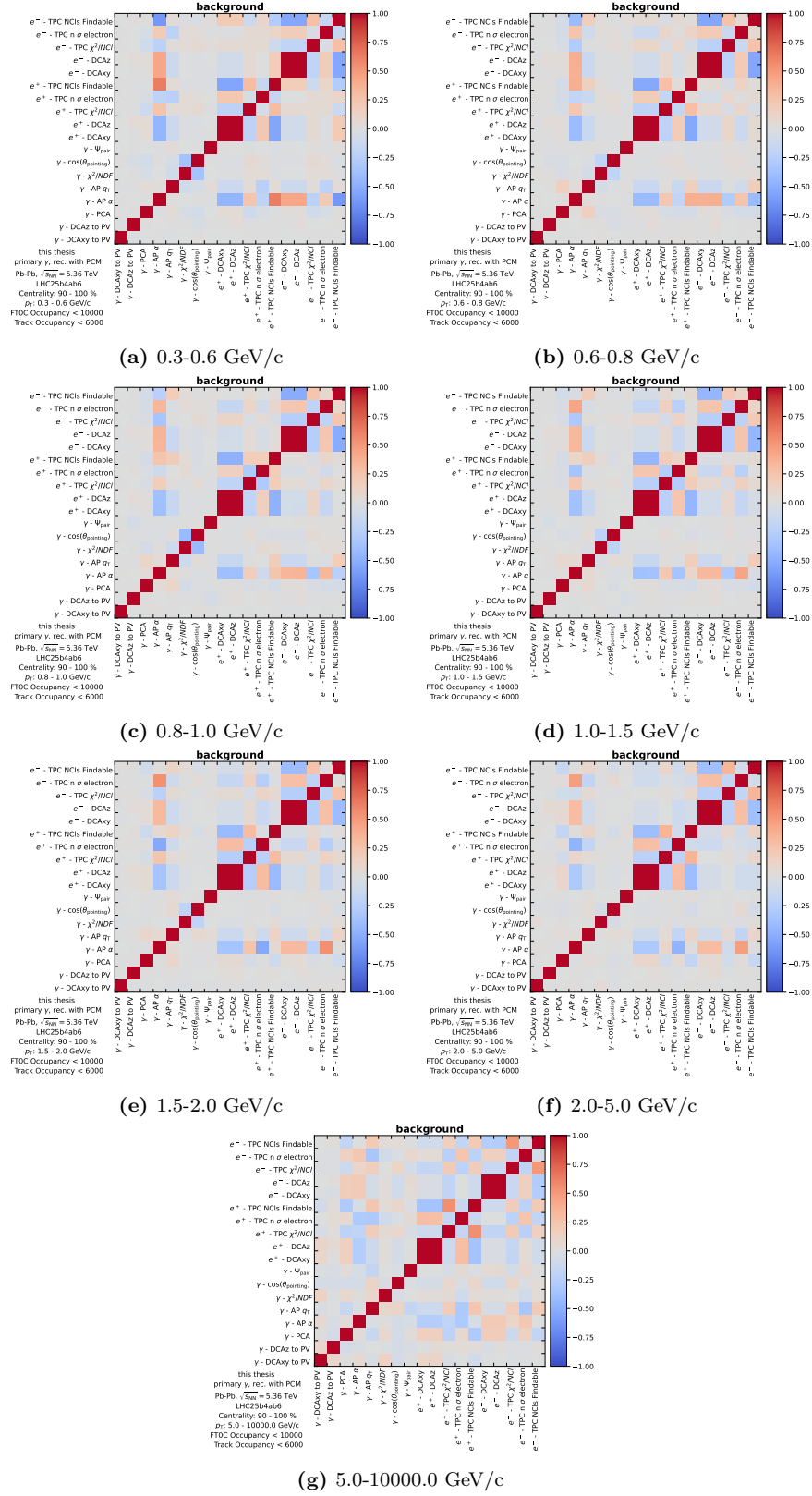


Figure 40: Correlation map of background candidates for all p_T bins in the centrality range 90-100 %